

logica_tia

Totally Integrated Automation Portal		
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factoryio [CPU 1511-1 PN]

factoryio

General\Project information

Name	factoryio	Author	Andrei
Comment		Rack	0
Slot	1		

General\Catalog information

Short designation	CPU 1511-1 PN	Description	CPU with display; work memory 150 KB code and 1 MB data; 60 ns bit operation time; 4-stage protection concept, technology functions: motion control, closed-loop control, counting & measuring; tracing; PROFINET IO controller, supports RT/IRT, performance upgrade PROFINET V2.3, 2 ports, I-device, MRP, MRPD, transport protocol TCP/IP, secure Open User Communication, S7 communication, Web server, DNS client, OPC UA server data access, constant bus cycle time, routing; Runtime options, firmware V2.1
Article number	6ES7 511-1AK01-0AB0	Firmware version	V2.1
	False		

General\Identification & Maintenance

Plant designation		Location identifier	
Installation date	2016-04-07 13:20:53.143	Additional information	

General\Checksums

Text lists	FA 70 E8 75 1D 5A 8E 29	Software	07 AE 60 E1 17 B3 C1 61
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Connection resources\

	Station resources - Reserved - Maximum	Station resources - Reserved - Configured	Station resources - Dynamic - Configured	Module resources - factoryio [CPU 1511-1 PN] - Configured
Maximum number of resources:		10	54	64
	Maximum	Configured	Configured	Configured
PG communication:	4	-	-	-
HMI communication:	4	2	0	2
S7 communication:	0	-	0	0
Open user communication:	0	-	0	0
Web communication:	2	-	-	-
Other communication:	-	-	0	0
Total resources used:		2	0	2
Available resources:		8	54	62

Overview of addresses\Overview of addresses\Overview of addresses

Inputs	True	Outputs	True
Address gaps	False	Slot	True

Totally Integrated Automation Portal							
Type	I	Addr. from	0	Addr. to	15	Module	AI 8xU//RTD/TC ST_1
PIP	None	OB	-	Device name	factoryio [CPU 1511-1 PN]	Device number	-
Size	16 Bytes	Master / IO-system	-	Rack	0	Slot	2
Type	I	Addr. from	16	Addr. to	31	Module	AI 8xU//RTD/TC ST_2
PIP	None	OB	-	Device name	factoryio [CPU 1511-1 PN]	Device number	-
Size	16 Bytes	Master / IO-system	-	Rack	0	Slot	3
Type	I	Addr. from	32	Addr. to	47	Module	AI 8xU//RTD/TC ST_3
PIP	None	OB	-	Device name	factoryio [CPU 1511-1 PN]	Device number	-
Size	16 Bytes	Master / IO-system	-	Rack	0	Slot	4
Type	I	Addr. from	48	Addr. to	63	Module	AI 8xU//RTD/TC ST_4
PIP	None	OB	-	Device name	factoryio [CPU 1511-1 PN]	Device number	-
Size	16 Bytes	Master / IO-system	-	Rack	0	Slot	5
Type	I	Addr. from	512	Addr. to	513	Module	DI 16x24VDC BA_1
PIP	None	OB	-	Device name	factoryio [CPU 1511-1 PN]	Device number	-
Size	2 Bytes	Master / IO-system	-	Rack	0	Slot	6

Totally Integrated Automation Portal

factoryio [CPU 1511-1 PN] / Program blocks

Main [OB1]

Main Properties

General

Name	Main	Number	1	Type	OB
Language	FBD	Numbering	Automatic		

Information

Title	"Main Program Sweep (Cycle)"	Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Default value
▼ Input		
Initial_Call	Bool	
Remanence	Bool	
Temp		
Constant		

Network 1:

%FC1

"MHJ-PLC-Lab-Function-S71500"

... — EN

ENO —

Network 2:

%DB3

"Blue_DB"

%FB1

"Blue"

... — EN

ENO —

Network 3:

%DB5

"Green_DB"

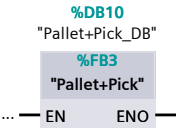
%FB2

"Green"

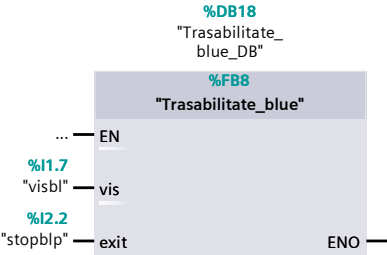
... — EN

ENO —

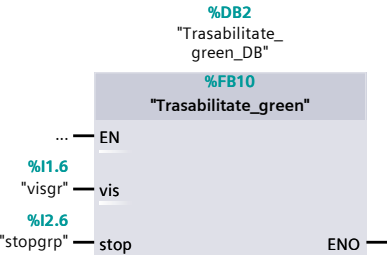
Network 4:



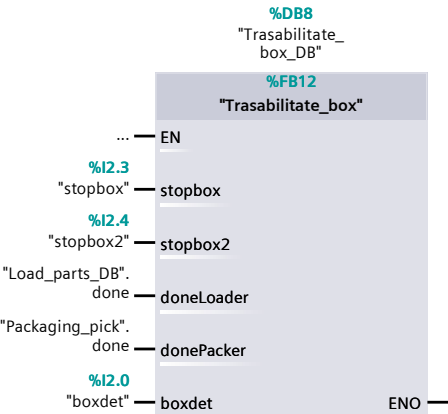
Network 5:



Network 6:



Network 7:



Network 8:

factoryio [CPU 1511-1 PN] / Program blocks

MHJ-PLC-Lab-Function-S71500 [FC1]

MHJ-PLC-Lab-Function-S71500 Properties					
General					
Name	MHJ-PLC-Lab-Function-S71500	Number	1	Type	FC
Language	SCL	Numbering	Automatic		
Information					
Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Default value
Input		
Output		
InOut		
▼ Temp		
Value	Byte	
ForCounter	Int	
▼ Constant		
Value_01_DW	DWord	16#1223_5486
Value_02_DW	DWord	16#A6C9_D1F5
▼ Return		
MHJ-PLC-Lab-Function-S71500	Void	

```
0001
0002 #Value:=PEEK(area := 16#82,
0003     dbNumber := 0,
0004     byteOffset := 511);
0005 #Value := #Value + 1;
0006
0007 POKE(area := 16#82,
0008     dbNumber := 0,
0009     byteOffset := 511,
0010     value := #Value);
0011
0012 POKE(area := 16#82,
0013     dbNumber := 0,
0014     byteOffset := 1016,
0015     value := #Value_01_DW);
0016 POKE(area := 16#82,
0017     dbNumber := 0,
0018     byteOffset := 1020,
0019     value := #Value_02_DW);
0020 FOR #ForCounter := 0 TO 63 DO
0021     #Value := PEEK(area := 16#1,
0022         dbNumber := 0,
0023         byteOffset := #ForCounter);
0024     POKE(area := 16#81,
0025         dbNumber := 0,
0026         byteOffset := #ForCounter,
0027         value := #Value);
0028 END_FOR;
0029 #Value := PEEK(area := 16#1,
```

Totally Integrated Automation Portal		
<pre>0030 dbNumber := 0, 0031 byteOffset := 512); 0032 POKE(area := 16#82, 0033 dbNumber := 0, 0034 byteOffset := 512, 0035 value := #Value); 0036</pre>		

factoryio [CPU 1511-1 PN] / Program blocks

Pick_simplu_1 [FB9]

Pick_simplu_1 Properties

General

Name	Pick_simplu_1	Number	9	Type	FB
Language	SCL	Numbering	Automatic		

Information

Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Default value	Retain
▼ Input			
enable	Bool	false	Non-retain
pxpos	Real	0.0	Non-retain
pzpos	Real	0.0	Non-retain
xp	Real	0.0	Non-retain
zp	Real	0.0	Non-retain
xd	Real	0.0	Non-retain
zd	Real	0.0	Non-retain
pboxdet	Bool	false	Non-retain
▼ Output			
pxsetp	Real	0.0	Non-retain
pzsetp	Real	0.0	Non-retain
pgrab	Bool	false	Non-retain
done	Bool	false	Non-retain
rotccw	Bool	false	Non-retain
rotcw	Bool	false	Non-retain
InOut			
▼ Static			
inst_Timer	IEC_TIMER		Non-retain
delay_active	Bool	false	Non-retain
old_step	DInt	0	Non-retain
step	DInt	0	Non-retain
Temp			
Constant			

```
0001 #inst_Timer.TON(IN := #delay_active,
0002     PT := T#800ms);
0003
0004 CASE #step OF
0005   0: // IDLE
0006     #pgrab := FALSE;
0007     #rotccw := FALSE;
0008     #rotcw := FALSE;
0009     #done := FALSE;
0010     IF #enable THEN
0011       #step := 10;
0012     END_IF;
0013
0014   10:
0015     #pxsetp := 1.7;
0016     #pzsetp := 6.4;
```

Totally Integrated Automation Portal		
0017	#delay_active := TRUE;	
0018	IF #inst_Timer.Q THEN	
0019	#delay_active := FALSE;	
0020	#pgrab := TRUE;	
0021	#step := 20;	
0022	END_IF;	
0023		
0024	20:	
0025	#delay_active := TRUE;	
0026	IF #inst_Timer.Q THEN	
0027	#delay_active := FALSE;	
0028	#step := 30;	
0029	END_IF;	
0030		
0031	30:	
0032	#pzsetp := 0.0;	
0033	#pxsetp := 1.3;	
0034	#delay_active := TRUE;	
0035	IF #inst_Timer.Q THEN	
0036	#delay_active := FALSE;	
0037	#step := 40;	
0038	END_IF;	
0039		
0040	40: // ROT	
0041	#rotcw := TRUE;	
0042	#delay_active := TRUE;	
0043	IF #inst_Timer.Q THEN	
0044	#delay_active := FALSE;	
0045	#rotcw := FALSE; // coboară semnalul	
0046	#step := 41;	
0047	END_IF;	
0048		
0049	41:	
0050	#rotcw := FALSE;	
0051	#delay_active := TRUE;	
0052	IF #inst_Timer.Q THEN	
0053	#delay_active := FALSE;	
0054	#step := 42;	
0055	END_IF;	
0056		
0057	42: // ROT	
0058	#rotcw := TRUE;	
0059	#delay_active := TRUE;	
0060	IF #inst_Timer.Q THEN	
0061	#delay_active := FALSE;	
0062	#rotcw := FALSE;	
0063	#step := 50;	
0064	END_IF;	
0065		
0066	50: // DROP	
0067	#pgrab := FALSE;	
0068	#delay_active := TRUE;	
0069	IF #inst_Timer.Q THEN	
0070	#delay_active := FALSE;	
0071	#step := 60;	
0072	END_IF;	
0073		
0074	60: //HOME - rotCCW	
0075	#pxsetp := 0.0;	

Totally Integrated Automation Portal		
<pre>0076 #pzsetp := 0.0; 0077 #rotccw := TRUE; 0078 #delay_active := TRUE; 0079 IF #inst_Timer.Q THEN 0080 #delay_active := FALSE; 0081 #rotccw := FALSE; 0082 #step := 61; 0083 END_IF; 0084 0085 61: 0086 #rotccw := FALSE; 0087 #delay_active := TRUE; 0088 IF #inst_Timer.Q THEN 0089 #delay_active := FALSE; 0090 #step := 62; 0091 END_IF; 0092 0093 62: // ROT CCW 2 0094 #rotccw := TRUE; 0095 #delay_active := TRUE; 0096 IF #inst_Timer.Q THEN 0097 #delay_active := FALSE; 0098 #rotccw := FALSE; 0099 #step := 70; 0100 END_IF; 0101 0102 70: 0103 #done := TRUE; 0104 0105 END_CASE; 0106 0107 // RESET 0108 IF NOT #enable THEN 0109 IF #step = 0 OR #step = 70 THEN 0110 #step := 0; 0111 #pgrab := FALSE; 0112 #rotcw := FALSE; 0113 #rotccw := FALSE; 0114 #delay_active := FALSE; 0115 END_IF; 0116 END_IF;</pre>		

factoryio [CPU 1511-1 PN] / Program blocks

Count_fct [FB7]

Count_fct Properties					
General					
Name	Count_fct	Number	7	Type	FB
Language	SCL	Numbering	Automatic		
Information					
Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Default value	Retain
▼ Input			
sensor	Bool	false	Non-retain
reset	Bool	false	Non-retain
PV	Int	0	Non-retain
▼ Output			
done	Bool	false	Non-retain
InOut			
▼ Static			
CV	Int	0	Non-retain
sensor_prev	Bool	false	Non-retain
Temp			
Constant			

```
0001
0002 IF #sensor AND NOT #sensor_prev THEN
0003     #CV := #CV + 1;
0004 END_IF;
0005
0006 #sensor_prev := #sensor;
0007
0008 IF #CV >= #PV AND #PV > 0 THEN
0009     #done := TRUE;
0010 ELSE
0011     #done := FALSE;
0012 END_IF;
0013
0014 IF #reset THEN
0015     #CV := 0;
0016     #done := FALSE;
0017 END_IF;
```

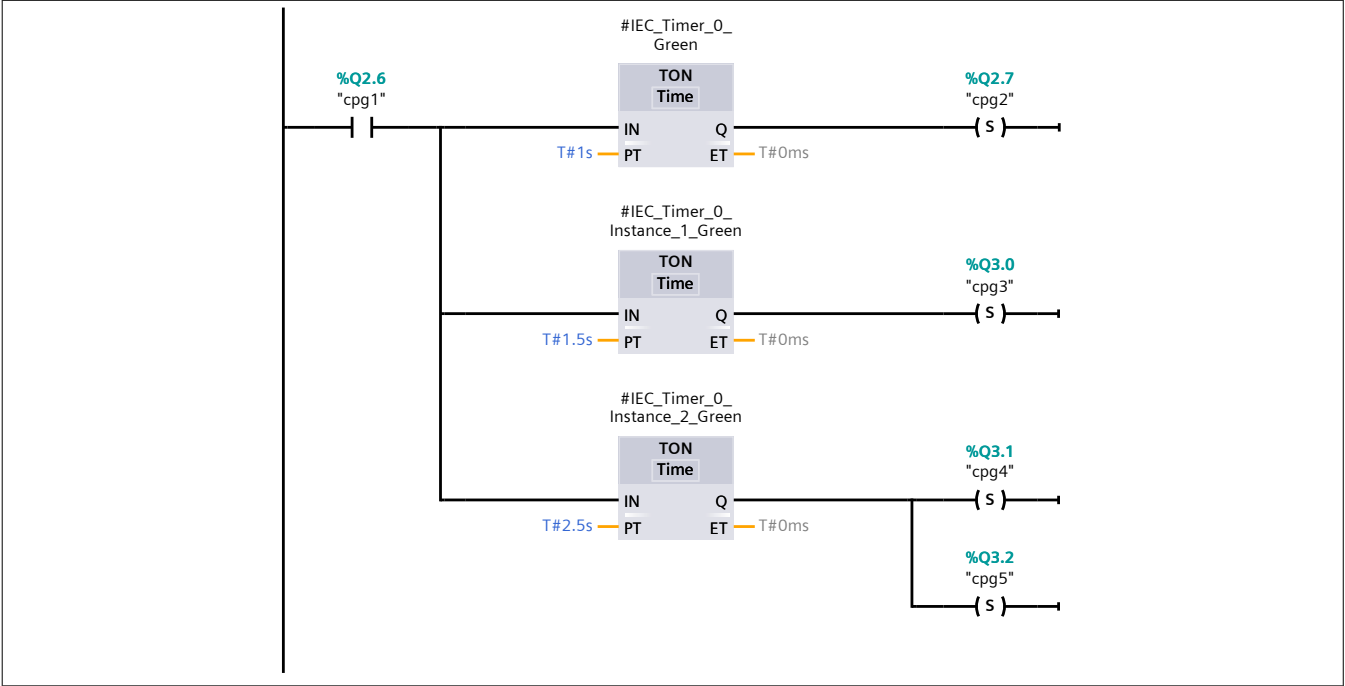
factoryio [CPU 1511-1 PN] / Program blocks

Green [FB2]

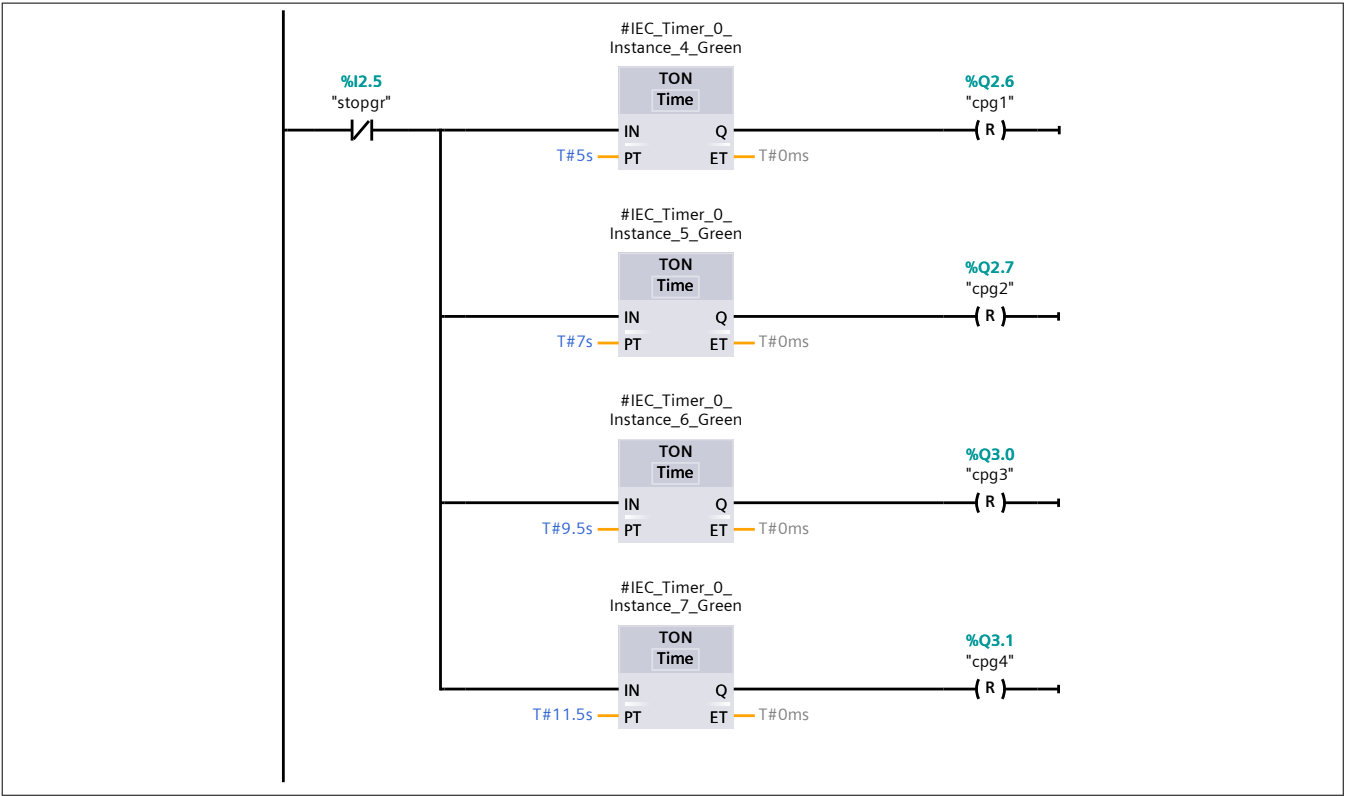
Green Properties					
General					
Name	Green	Number	2	Type	FB
Language	LAD	Numbering	Automatic		
Information					
Title		Author		Comment	Flux piesa verde de la raw material -> machining0 -> inspectie vision (a 10-a piesa mereu defecta) ->transport conveyoare pana la zona de asamblare
Family		Version	0.1	User-defined ID	

Name	Data type	Default value	Retain
Input			
Output			
InOut			
▼ Static			
IEC_Timer_0_Green	TON_TIME		Non-retain
IEC_Timer_0_Instance_1_Green	TON_TIME		Non-retain
IEC_Timer_0_Instance_2_Green	TON_TIME		Non-retain
IEC_Timer_0_Instance_4_Green	TON_TIME		Non-retain
IEC_Timer_0_Instance_5_Green	TON_TIME		Non-retain
IEC_Timer_0_Instance_6_Green	TON_TIME		Non-retain
IEC_Timer_0_Instance_7_Green	TON_TIME		Non-retain
IEC_Timer_0_Instance_8_Green	TON_TIME		Non-retain
IEC_Timer_0_Instance_10_Green	TON_TIME		Non-retain
Tlmer0_Blue_1	TON_TIME		Non-retain
Tlmer0_Blue_2	TON_TIME		Non-retain
Tlmer0_Blue_3	TON_TIME		Non-retain
Tlmer0_Blue_4	TON_TIME		Non-retain
Tlmer0_Blue_5	TON_TIME		Non-retain
Tlmer0_Blue_6	TON_TIME		Non-retain
Tlmer0_Blue_7	TON_TIME		Non-retain
Temp			
Constant			

Network 1: pornire conveyor raw



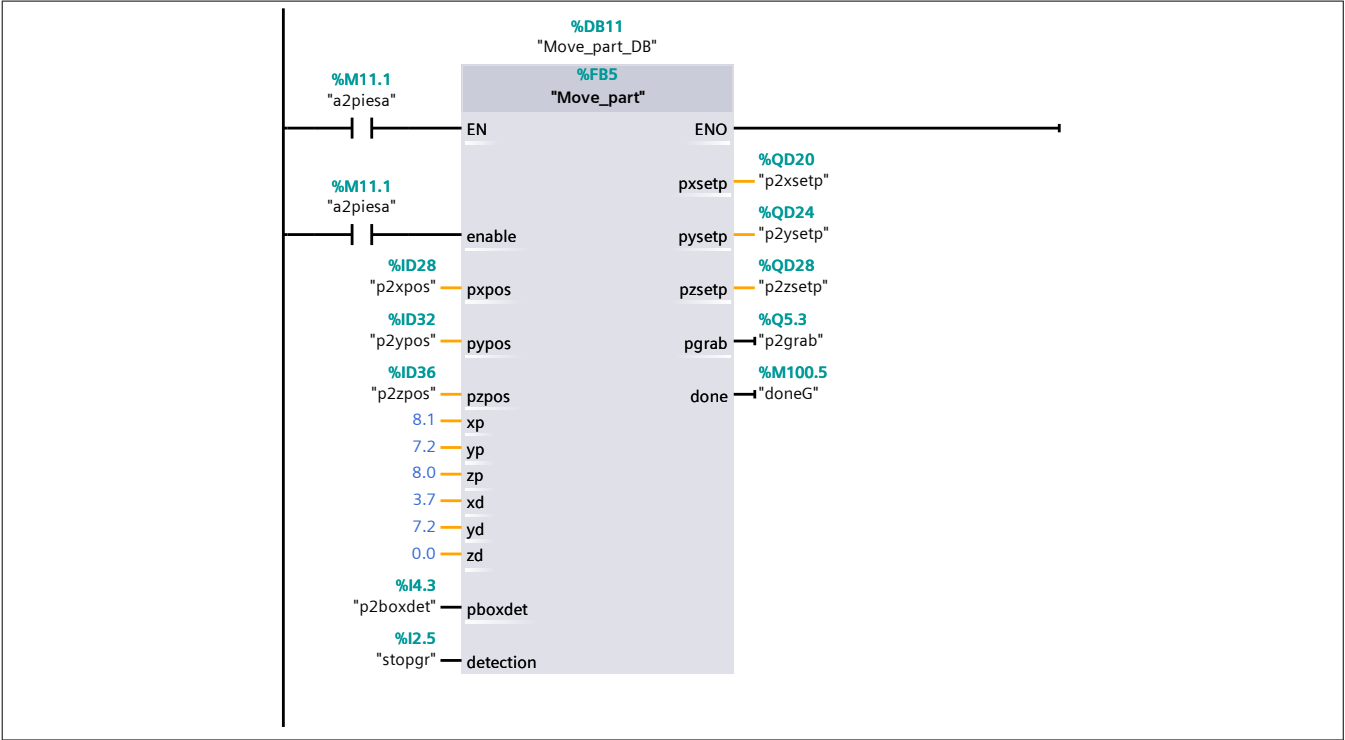
Network 6: resetare benzi



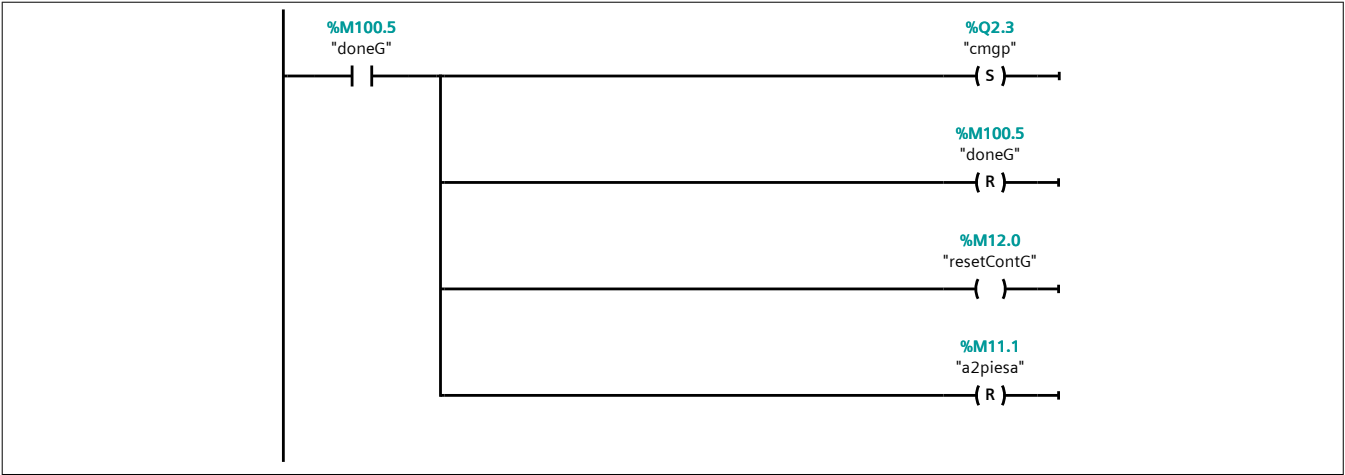
Network 7: oprire banda piesa rea



Network 8:



Network 9:



factoryio [CPU 1511-1 PN] / Program blocks

Load_parts [FB6]

Load_parts Properties

General

Name	Load_parts	Number	6	Type	FB
Language	SCL	Numbering	Automatic		

Information

Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Default value	Retain
▼ Input			
stopblp	Bool	false	Non-retain
stopgrp	Bool	false	Non-retain
boxdet	Bool	false	Non-retain
done_robot	Bool	false	Non-retain
stopbox	Bool	false	Non-retain
▼ Output			
xp	Real	0.0	Non-retain
yp	Real	0.0	Non-retain
zp	Real	0.0	Non-retain
xd	Real	0.0	Non-retain
yd	Real	0.0	Non-retain
zd	Real	0.0	Non-retain
done	Bool	false	Non-retain
InOut			
▼ Static			
step	DInt	0	Non-retain
counter	Int	0	Non-retain
is_box	Bool	false	Non-retain
enable_robot	Bool	false	Non-retain
inst_Timer	IEC_TIMER		Non-retain
delay_active	Bool	false	Non-retain
need_blue	Bool	false	Non-retain
Temp			
Constant			

```
0001 #inst_Timer.TON(IN := #delay_active,
0002 PT := T#2000ms);
0003 CASE #step OF
0004
0005 0: // IDLE
0006     #done := FALSE;
0007     #counter := 0;
0008     #enable_robot := FALSE;
0009
0010 IF #stopbox THEN
0011     #is_box := #boxdet;
0012     #step := 10;
0013 END_IF;
0014
0015 10: //
```

```

0016     IF #is_box THEN
0017         CASE #counter OF
0018             0: // albastru jos
0019                 #xp := 2.8;
0020                 #yp := 0.5;
0021                 #zp := 8.0;
0022                 #xd := 4.1;
0023                 #yd := 6.4;
0024                 #zd := 7.8;
0025                 #need_blue := TRUE;
0026             1: // verde
0027                 #xp := 6.7;
0028                 #yp := 0.5;
0029                 #zp := 8.0;
0030                 #xd := 4.1;
0031                 #yd := 6.4;
0032                 #zd := 6.6;
0033                 #need_blue := FALSE;
0034             2: // albastru
0035                 #xp := 2.8;
0036                 #yp := 0.5;
0037                 #zp := 8.0;
0038                 #xd := 4.1;
0039                 #yd := 6.4;
0040                 #zd := 5.4;
0041                 #need_blue := TRUE;
0042             3: // verde
0043                 #xp := 6.7;
0044                 #yp := 0.5;
0045                 #zp := 8.0;
0046                 #xd := 4.1;
0047                 #yd := 6.4;
0048                 #zd := 4.2;
0049                 #need_blue := FALSE;
0050         END_CASE;
0051     ELSE
0052         CASE #counter OF
0053             0: // palet verde jos
0054                 #xp := 6.7;
0055                 #yp := 0.5;
0056                 #zp := 8.0;
0057                 #xd := 2.8;
0058                 #yd := 6.4;
0059                 #zd := 6.1;
0060                 #need_blue := FALSE;
0061             1: // palet albastru sus
0062                 #xp := 2.7;
0063                 #yp := 0.5;
0064                 #zp := 8.0;
0065                 #xd := 2.8;
0066                 #yd := 6.4;
0067                 #zd := 4.8;
0068                 #need_blue := TRUE;
0069             2: // palet dreapta albastru jos
0070                 #xp := 2.7;
0071                 #yp := 0.5;
0072                 #zp := 8.0;
0073                 #xd := 5.0;
0074                 #yd := 6.4;

```

```

0075         #zd := 6.1;
0076         #need_blue := TRUE;
0077     3: // palet dreapta verde sus
0078         #xp := 6.7;
0079         #yp := 0.5;
0080         #zp := 8.0;
0081         #xd := 5.0;
0082         #yd := 6.4;
0083         #zd := 4.8;
0084         #need_blue := FALSE;
0085     END_CASE;
0086 END_IF;
0087 #step := 20;
0088
0089 20:
0090     IF #need_blue AND #stopblp THEN
0091         #enable_robot := TRUE;
0092         #step := 30;
0093     ELSIF NOT #need_blue AND #stopgrp THEN
0094         #enable_robot := TRUE;
0095         #step := 30;
0096     END_IF;
0097
0098 30:
0099     IF #done_robot THEN
0100         #enable_robot := FALSE;
0101         #counter := #counter + 1;
0102         #step := 40;
0103     END_IF;
0104
0105 40:
0106     IF #counter >= 4 THEN
0107         #done := TRUE;
0108         #step := 50;
0109     ELSE
0110         #step := 10;
0111     END_IF;
0112
0113 50:
0114     #done := TRUE;
0115
0116
0117     IF NOT #stopbox THEN
0118         #step := 0;
0119         #done := FALSE;
0120         #counter := 0;
0121     END_IF;
0122
0123 END_CASE;
0124
0125

```

factoryio [CPU 1511-1 PN] / Program blocks

Move_part [FB5]

Move_part Properties

General

Name	Move_part	Number	5	Type	FB
Language	SCL	Numbering	Automatic		

Information

Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Default value	Retain
▼ Input			
enable	Bool	false	Non-retain
pxpos	Real	0.0	Non-retain
pypos	Real	0.0	Non-retain
pzpos	Real	0.0	Non-retain
xp	Real	0.0	Non-retain
yp	Real	0.0	Non-retain
zp	Real	0.0	Non-retain
xd	Real	0.0	Non-retain
yd	Real	0.0	Non-retain
zd	Real	0.0	Non-retain
pboxdet	Bool	false	Non-retain
detection	Bool	false	Non-retain
▼ Output			
pxsetp	Real	0.0	Non-retain
pysetp	Real	0.0	Non-retain
pzsetp	Real	0.0	Non-retain
pgrab	Bool	false	Non-retain
done	Bool	false	Non-retain
InOut			
▼ Static			
inst_Timer	IEC_TIMER		Non-retain
delay_active	Bool	false	Non-retain
step	DInt	0	Non-retain
Temp			
Constant			

```
0001 #inst_Timer.TON(IN := #delay_active,
0002     PT := T#1000ms);
0003
0004 CASE #step OF
0005   0: // home
0006     #pxsetp := 0.0;
0007     #pysetp := 0.0;
0008     #pzsetp := 0.0;
0009     #pgrab := FALSE;
0010     IF #enable THEN
0011       #done := FALSE;
0012       #step := 5;
0013     END_IF;
0014
```

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0015	5: //Z 0	
0016	#pzsetp := 0.0;	
0017	IF ABS(#pzpos - 0.0) < 0.2 THEN	
0018	#step := 10;	
0019	END_IF;	
0020		
0021	10: //PICK (X, Y)	
0022	#pxsetp := #xp;	
0023	#pysetp := #yp;	
0024	#pzsetp := 0.0;	
0025		
0026	IF ABS(#pypos - #yp) < 0.3 AND ABS(#pxpos - #xp) < 0.3 THEN	
0027	#step := 15;	
0028	END_IF;	
0029		
0030	15:	
0031	#pzsetp := #zp;	
0032	IF ABS(#pzpos - #zp) < 0.1 THEN	
0033	#delay_active := TRUE;	
0034	IF #inst_Timer.Q THEN	
0035	#delay_active := FALSE;	
0036	#step := 20;	
0037	END_IF;	
0038	END_IF;	
0039		
0040	20: // prinde	
0041	#pgrab := TRUE;	
0042	#delay_active := TRUE;	
0043	IF #inst_Timer.Q THEN	
0044	#delay_active := FALSE;	
0045	#step := 30;	
0046	END_IF;	
0047		
0048	30:	
0049	#pzsetp := 0.0;	
0050	IF ABS(#pzpos - 0.0) < 0.2 THEN	
0051	#delay_active := TRUE;	
0052	IF #inst_Timer.Q THEN	
0053	#delay_active := FALSE;	
0054	#step := 35;	
0055	END_IF;	
0056	END_IF;	
0057		
0058	35: // drop (X, Y)	
0059	#pxsetp := #xd;	
0060	#pysetp := #yd;	
0061	IF ABS(#pxpos - #xd) < 0.2 AND ABS(#pypos - #yd) < 0.2 THEN	
0062	#step := 40;	
0063	END_IF;	
0064		
0065	40:	
0066	#pzsetp := #zd;	
0067	IF ABS(#pzpos - #zd) < 0.1 THEN	
0068	#step := 45;	
0069	END_IF;	
0070		
0071	45: // release	
0072	#pgrab := FALSE;	
0073	#delay_active := TRUE;	

Totally Integrated Automation Portal		
<pre>0074 IF #inst_Timer.Q THEN 0075 #delay_active := FALSE; 0076 #step := 47; 0077 END_IF; 0078 0079 47: 0080 #pzsetp := 0.0; 0081 #pxsetp := #xd; 0082 #pysetp := #yd; 0083 0084 0085 IF ABS(#pzpos - 0.0) < 0.2 THEN 0086 #step := 50; 0087 END_IF; 0088 0089 50: // done reset 0090 #done := TRUE; 0091 #step := 0; // 0092 END_CASE; 0093 0094 // reset fortat 0095 IF NOT #enable THEN 0096 #step := 0; 0097 #pgrab := FALSE; 0098 #delay_active := FALSE; 0099 #done := FALSE; 0100 END_IF;</pre>		

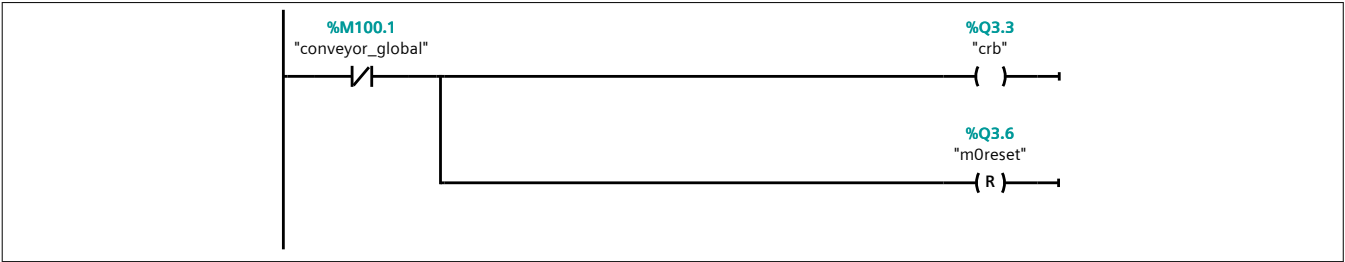
factoryio [CPU 1511-1 PN] / Program blocks

Blue [FB1]

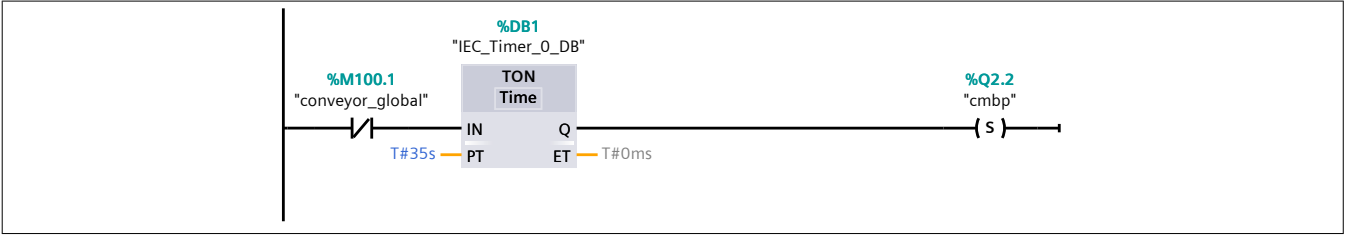
Blue Properties					
General					
Name	Blue	Number	1	Type	FB
Language	LAD	Numbering	Automatic		
Information					
Title		Author		Comment	Flux piesa albastra de la raw material -> machining0 -> inspectie vision (a 9-a piesa mereu defecta) ->transport conveyoare pana la zona de asamblare
Family		Version	0.1	User-defined ID	

Name	Data type	Default value	Retain
Input			
Output			
InOut			
▼ Static			
IEC_Timer_0_Instance	TON_TIME		Non-retain
IEC_Timer_0_Instance_1	TON_TIME		Non-retain
IEC_Timer_0_Instance_2	TON_TIME		Non-retain
IEC_Timer_0_Instance_3	TON_TIME		Non-retain
IEC_Timer_0_Instance_4	TON_TIME		Non-retain
IEC_Timer_0_Instance_5	TON_TIME		Non-retain
IEC_Timer_0_Instance_6	TON_TIME		Non-retain
IEC_Timer_0_Instance_7	TON_TIME		Non-retain
IEC_Timer_0_Instance_8	TON_TIME		Non-retain
IEC_Timer_0_Instance_9	TON_TIME		Non-retain
IEC_Timer_0_Instance_10	TON_TIME		Non-retain
count_blue	CTU_INT		Retain
timer_pick	TON_TIME		Non-retain
Temp			
Constant			

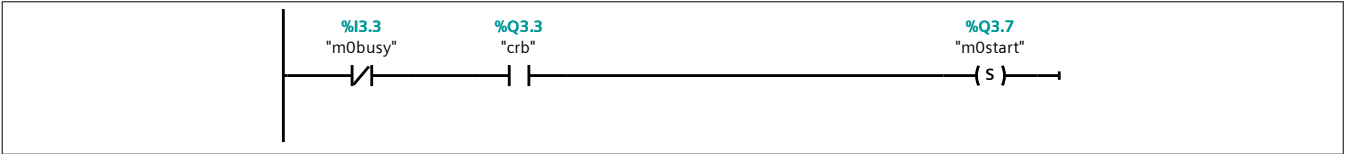
Network 1: pornire conveyor raw



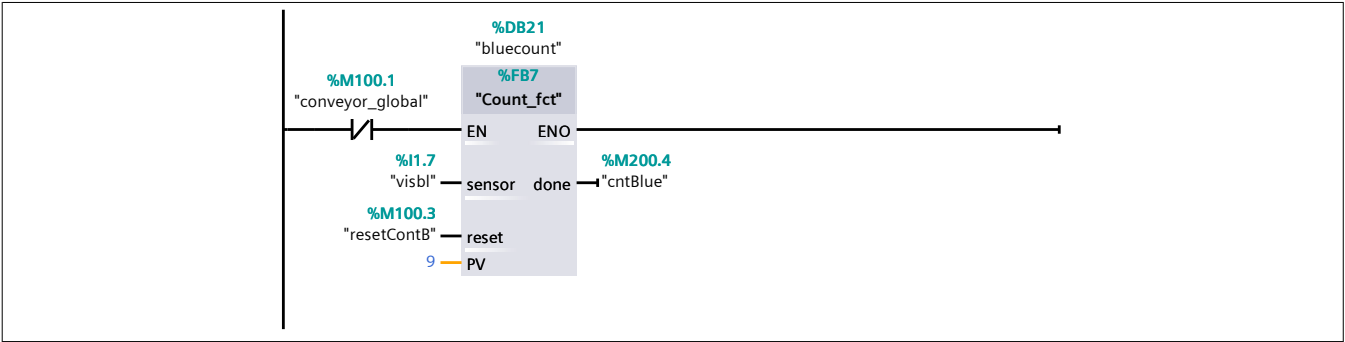
Network 2: pornire cmbp a.i. sa porneasca cu aprox. 10-15s inainte ca prima piesa sa fie gata



Network 3: start machining 1



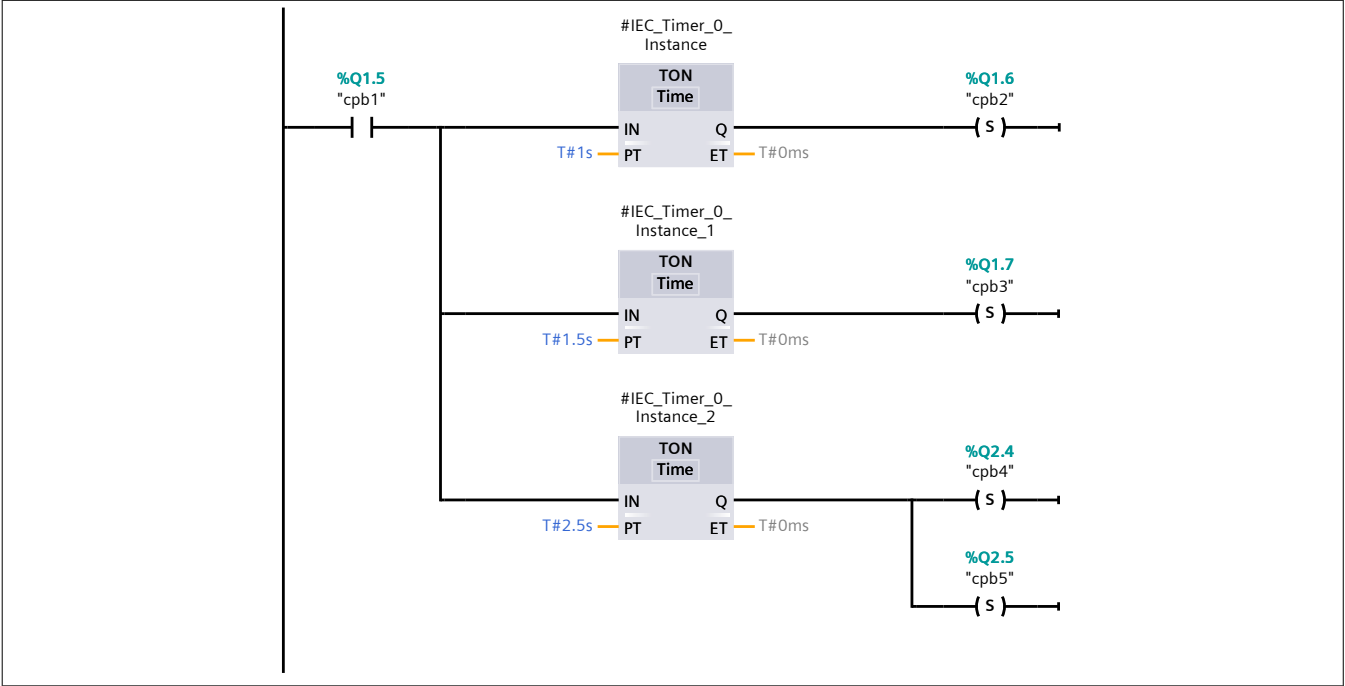
Network 4:



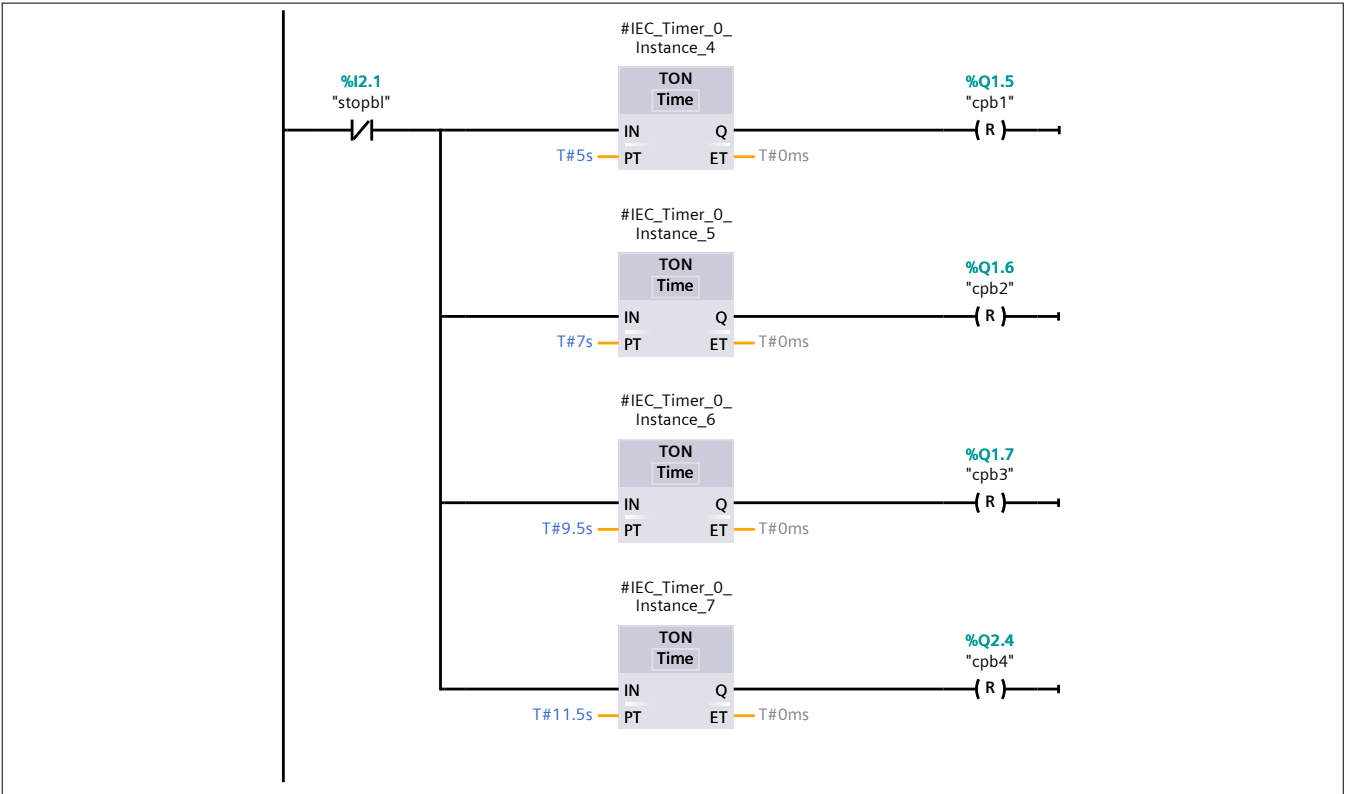
Network 5:



Network 6: pornire succesiva benzi



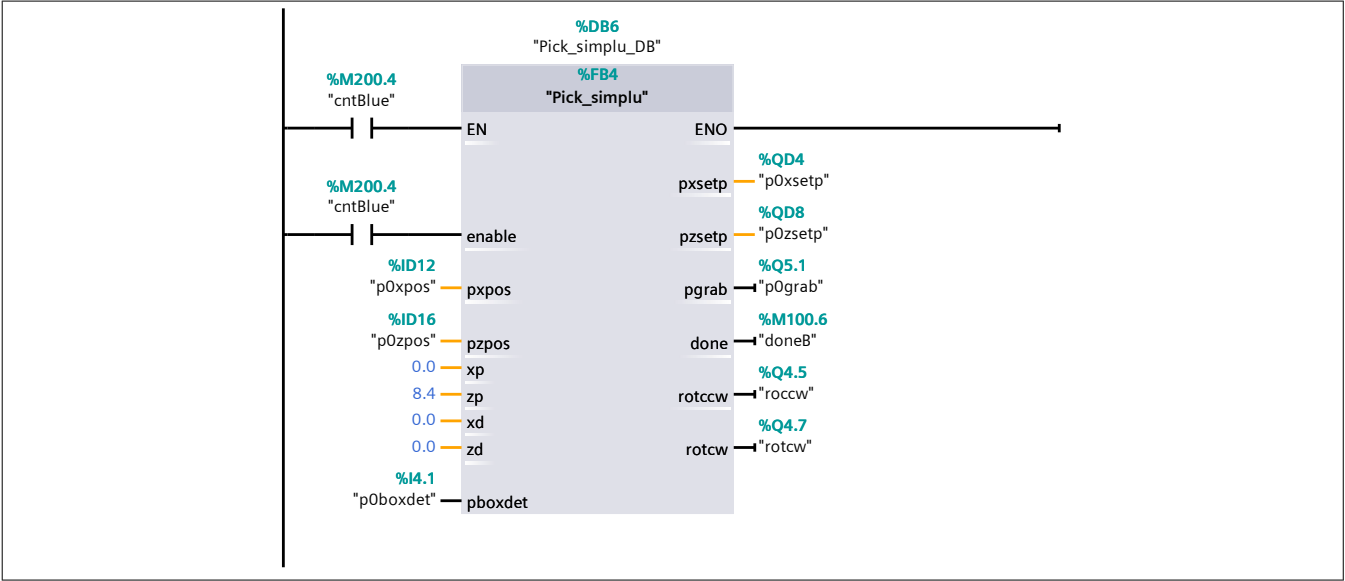
Network 7: resettare benzi



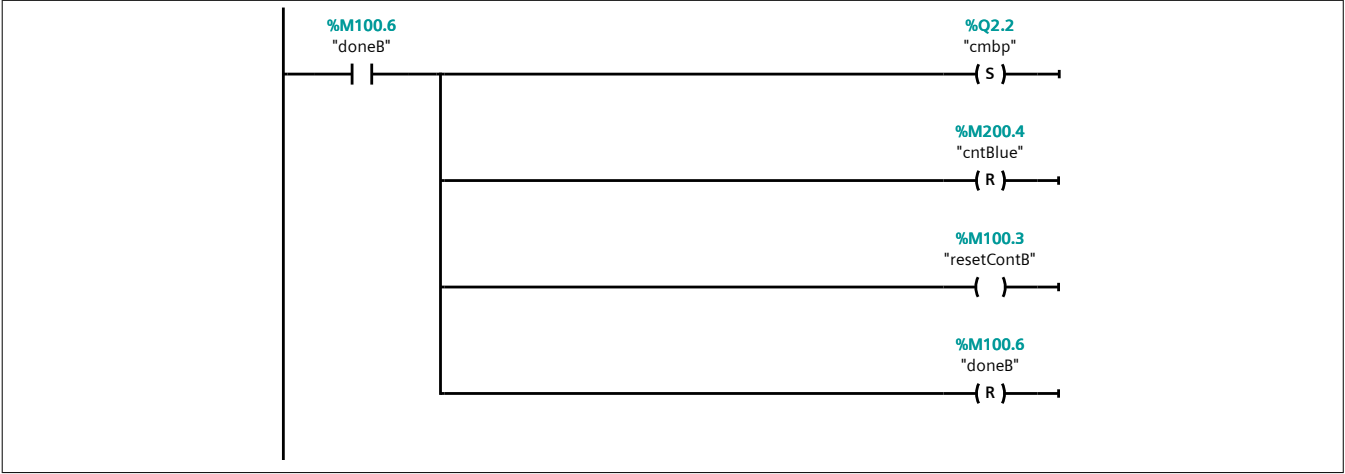
Network 8: oprire banda piesa rea



Network 9: eliminare piesa defecta



Network 10:



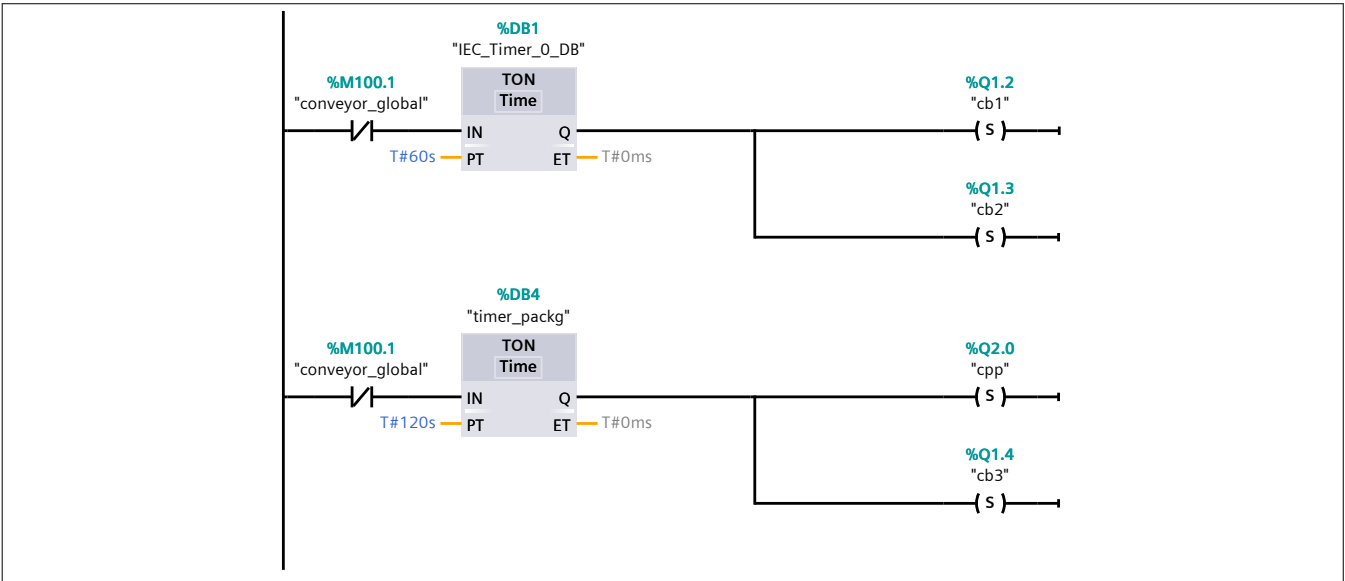
factoryio [CPU 1511-1 PN] / Program blocks

Pallet+Pick [FB3]

Pallet+Pick Properties					
General					
Name	Pallet+Pick	Number	3	Type	FB
Language	LAD	Numbering	Automatic		
Information					
Title		Author		Comment	Blocare conbeyoare, logi- ca de pornire a cutiei/pa- let si operatia de pick and place aferenta
Family		Version	0.1	User-defined ID	

Name	Data type	Default value	Retain
Input			
Output			
InOut			
Static			
Temp			
Constant			

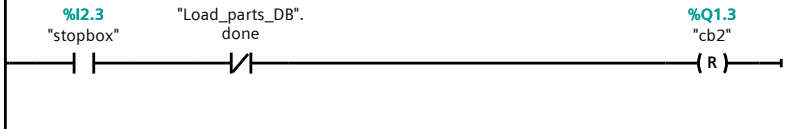
Network 1: pornire conveyoare paleti/cutie



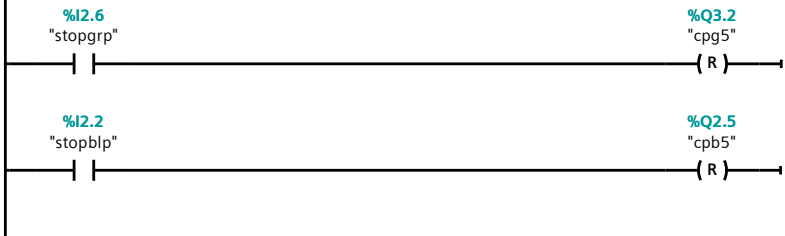
Network 2: cutie sau palet?



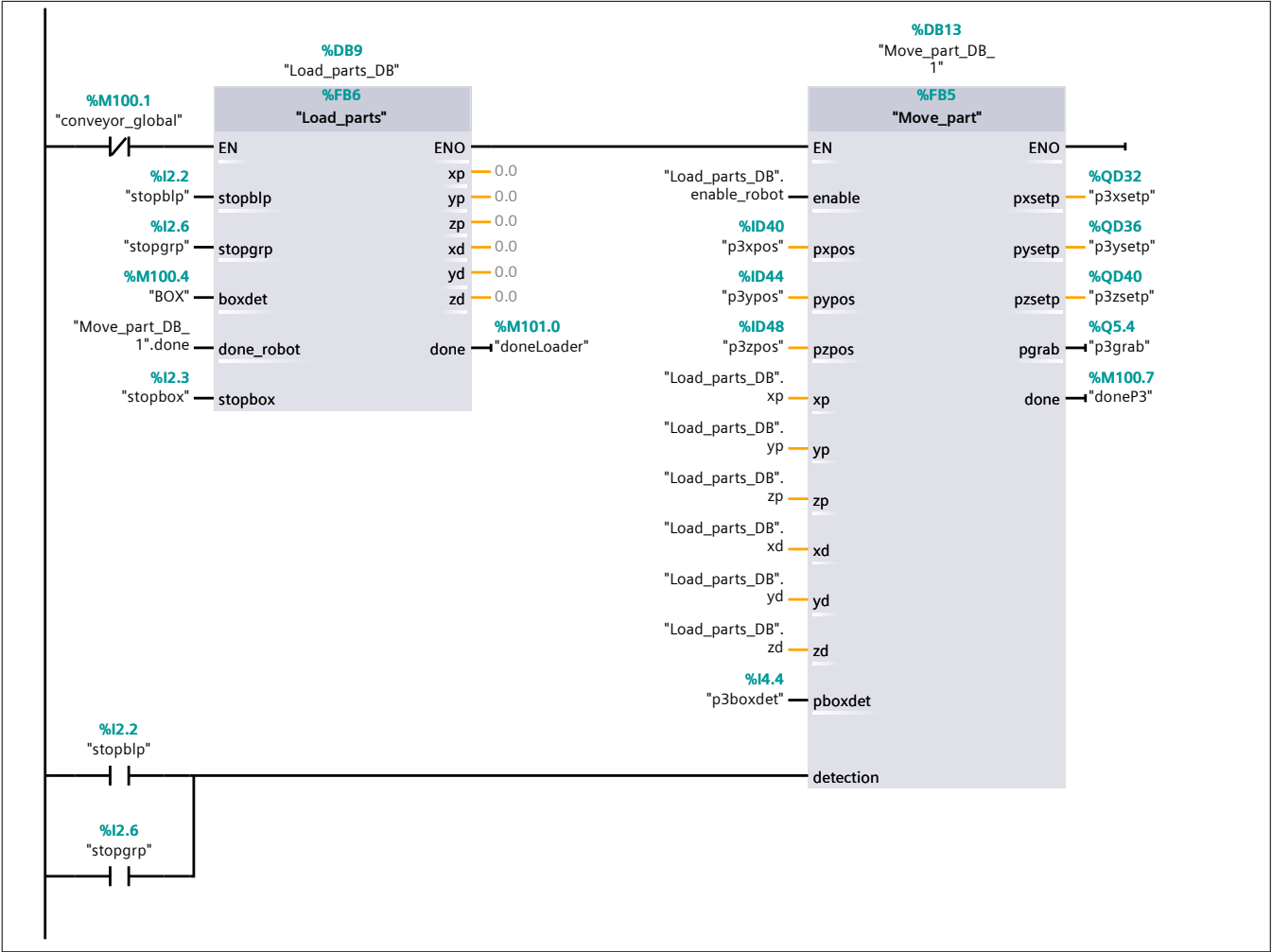
Network 3: oprire conveyor pentru pick



Network 4:

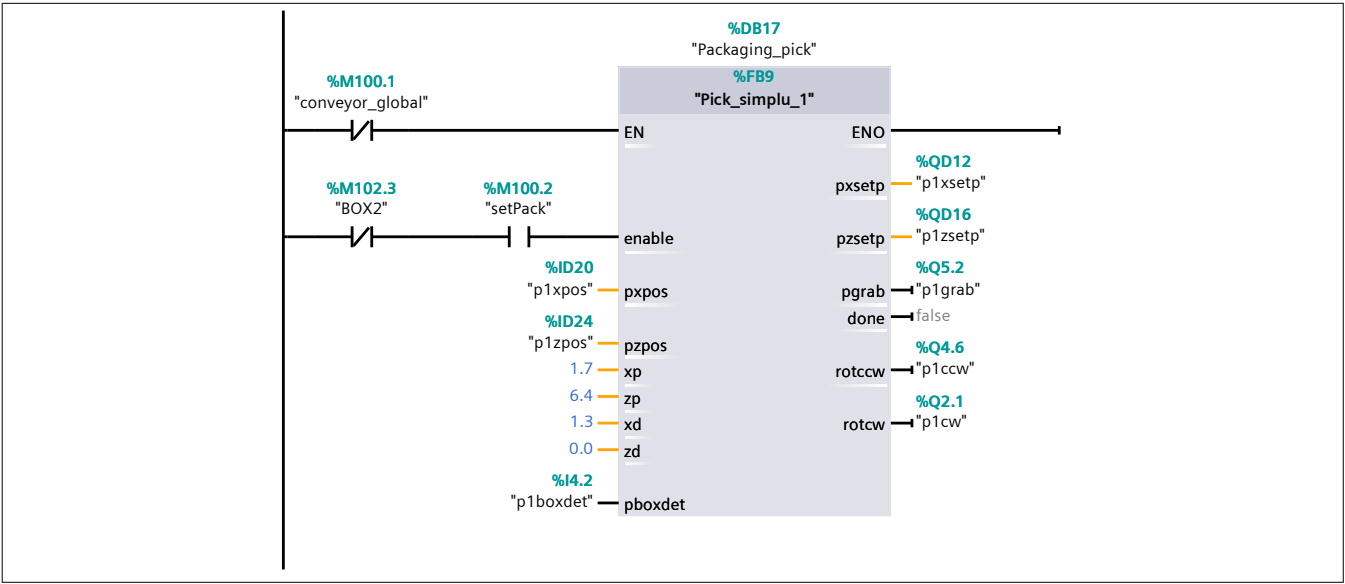


Network 5: algoritm punere piese + pick&place

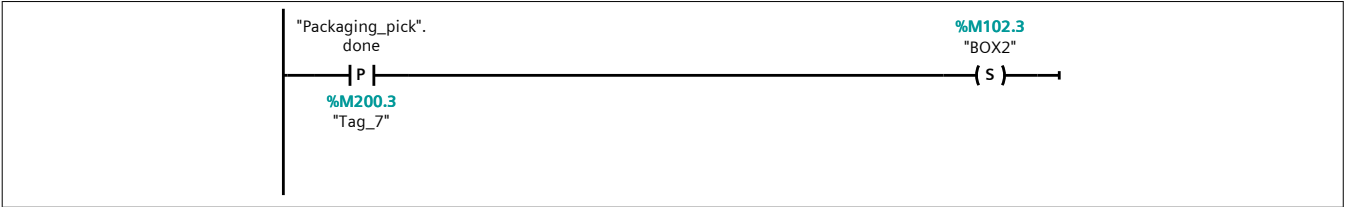


Network 6: resetari pentru reincepere ciclu

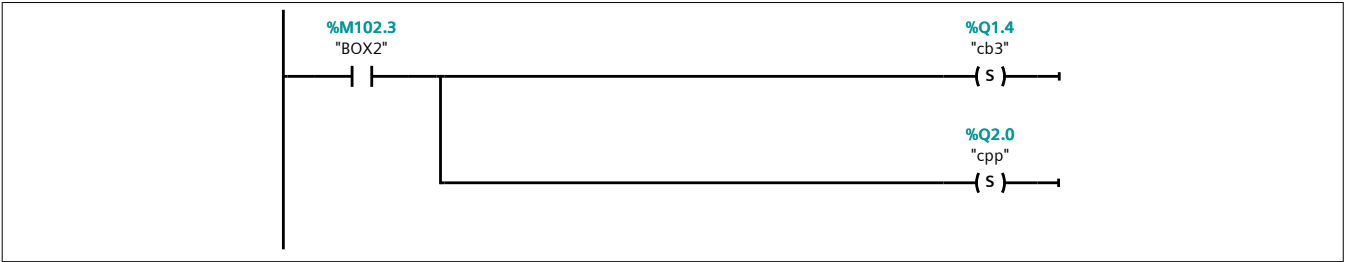
Totally Integrated Automation Portal		
"Load_parts_DB".done %Q1.4 "cb3" (S) %M102.1 "palletpickM" %Q1.3 "cb2" (S) %M100.4 "BOX" (R) %M101.0 "doneLoader" (R) %M100.7 "doneP3" (R)		
Network 7: cp 5 merge cat timp nu este piesa la capat %I2.6 "stopgrp" %Q3.2 "cpg5" (S) %I2.2 "stopblp" %Q2.5 "cpb5" (S)		
Network 8: oprire banda pt grup si packaging %I2.4 "stopbox2" %Q1.4 "cb3" (R) %I2.7 "stoppack" %Q2.0 "cpp" (R)		
Network 9: set marker %I2.4 "stopbox2" %I2.7 "stoppack" %M100.2 "setPack" (S)		
Network 10: pick and place pt palet		



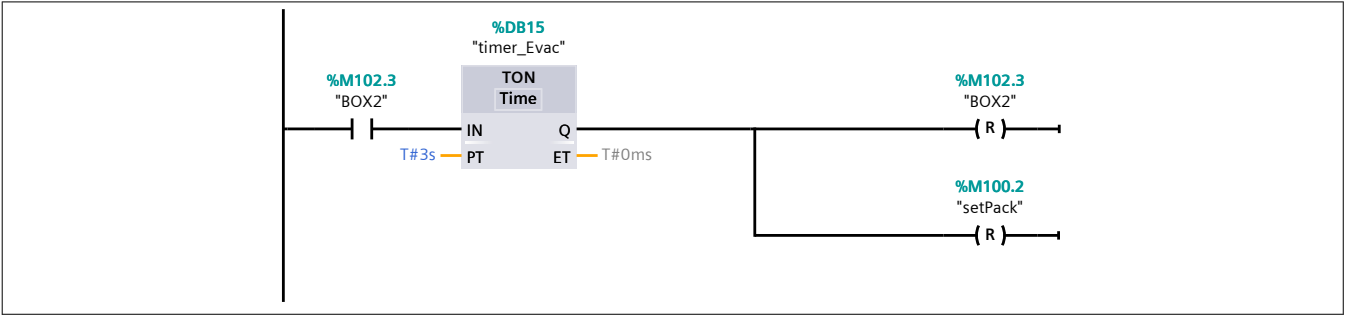
Network 11: set marker



Network 12: pornire conveyoare



Network 13: resetare dupa 3s



factoryio [CPU 1511-1 PN] / Program blocks

Pick_simplu [FB4]

Pick_simplu Properties

General

Name	Pick_simplu	Number	4	Type	FB
Language	SCL	Numbering	Automatic		

Information

Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Default value	Retain
▼ Input			
enable	Bool	false	Non-retain
pxpos	Real	0.0	Non-retain
pzpos	Real	0.0	Non-retain
xp	Real	0.0	Non-retain
zp	Real	0.0	Non-retain
xd	Real	0.0	Non-retain
zd	Real	0.0	Non-retain
pboxdet	Bool	false	Non-retain
▼ Output			
pxsetp	Real	0.0	Non-retain
pzsetp	Real	0.0	Non-retain
pgrab	Bool	false	Non-retain
done	Bool	false	Non-retain
rotccw	Bool	false	Non-retain
rotcw	Bool	false	Non-retain
InOut			
▼ Static			
inst_Timer	IEC_TIMER		Non-retain
delay_active	Bool	false	Non-retain
old_step	DInt	0	Non-retain
step	DInt	0	Non-retain
Temp			
Constant			

```
0001 #inst_Timer.TON(IN := #delay_active,
0002     PT := T#800ms);
0003 CASE #step OF
0004 0: // IDLE
0005     #pgrab := FALSE;
0006     #rotccw := FALSE;
0007     #rotcw := FALSE;
0008     #done := FALSE;
0009     IF #enable THEN
0010         #step := 10;
0011     END_IF;
0012
0013 10:
0014     #pxsetp := 0.0;
0015     #pzsetp := 8.4;
0016     #delay_active := TRUE;
```

Totally Integrated Automation Portal		
0017	IF #inst_Timer.Q THEN	
0018	#delay_active := FALSE;	
0019	#pgrab := TRUE;	
0020	#step := 20;	
0021	END_IF;	
0022		
0023	20:	
0024	#delay_active := TRUE;	
0025	IF #inst_Timer.Q THEN	
0026	#delay_active := FALSE;	
0027	#step := 30;	
0028	END_IF;	
0029		
0030	30:	
0031	#pzsetp := 0.0;	
0032	#pxsetp := 0.0;	
0033	#delay_active := TRUE;	
0034	IF #inst_Timer.Q THEN	
0035	#delay_active := FALSE;	
0036	#step := 40;	
0037	END_IF;	
0038		
0039	40: // ROT	
0040	#rotcw := TRUE;	
0041	#delay_active := TRUE;	
0042	IF #inst_Timer.Q THEN	
0043	#delay_active := FALSE;	
0044	#rotcw := FALSE;	
0045	#step := 41;	
0046	END_IF;	
0047		
0048	41:	
0049	#rotcw := FALSE;	
0050	#delay_active := TRUE;	
0051	IF #inst_Timer.Q THEN	
0052	#delay_active := FALSE;	
0053	#step := 42;	
0054	END_IF;	
0055		
0056	42: // ROT	
0057	#rotcw := TRUE;	
0058	#delay_active := TRUE;	
0059	IF #inst_Timer.Q THEN	
0060	#delay_active := FALSE;	
0061	#rotcw := FALSE;	
0062	#step := 50;	
0063	END_IF;	
0064		
0065	50: // DROP	
0066	#pgrab := FALSE;	
0067	#delay_active := TRUE;	
0068	IF #inst_Timer.Q THEN	
0069	#delay_active := FALSE;	
0070	#step := 60;	
0071	END_IF;	
0072		
0073	60: //HOME rotCCW	
0074	#pxsetp := 0.0;	
0075	#pzsetp := 0.0;	

Totally Integrated Automation Portal		
<pre> 0076 #rotccw := TRUE; 0077 #delay_active := TRUE; 0078 IF #inst_Timer.Q THEN 0079 #delay_active := FALSE; 0080 #rotccw := FALSE; 0081 #step := 61; 0082 END_IF; 0083 0084 61: 0085 #rotccw := FALSE; 0086 #delay_active := TRUE; 0087 IF #inst_Timer.Q THEN 0088 #delay_active := FALSE; 0089 #step := 62; 0090 END_IF; 0091 0092 62: // ROT CCW 2 0093 #rotccw := TRUE; 0094 #delay_active := TRUE; 0095 IF #inst_Timer.Q THEN 0096 #delay_active := FALSE; 0097 #rotccw := FALSE; 0098 #step := 70; 0099 END_IF; 0100 0101 70: // DONE 0102 #done := TRUE; 0103 #step := 0; 0104 END_CASE; 0105 // RESET 0106 IF NOT #enable THEN 0107 #step := 0; 0108 #pgrab := FALSE; 0109 #rotcw := FALSE; 0110 #rotccw := FALSE; 0111 #delay_active := FALSE; 0112 0113 #pxsetp := 0.0; 0114 #pzsetp := 0.0; 0115 0116 #done := FALSE; 0117 END_IF; </pre>		

factoryio [CPU 1511-1 PN] / Program blocks

Trasabilitate_blue [FB8]

Trasabilitate_blue Properties					
General					
Name	Trasabilitate_blue	Number	8	Type	FB
Language	SCL	Numbering	Automatic		
Information					
Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Default value
▼ Input		
vis	Bool	false
exit	Bool	false
Output		
InOut		
▼ Static		
Blue	Array[0..50] of "udt_Blue_Trace"	
idx	Int	0
edge3	Bool	false
edge4	Bool	false
Time_status	Int	0
Time_deadline	DTL	DTL#1970-01-01-00:00:00
Actual_Time	DTL	DTL#1970-01-01-00:00:00
j	Int	0
k	Int	0
nr_conf	Int	0
nr_neconf	Int	0
Total_tsum_DINT	DInt	0
Diff_time	Time	T#0ms
av_sec	Real	0.0
m	Int	0
Temp		
Constant		

```
0001
0002 #Time_status := RD_LOC_T(OUT => #Actual_Time);
0003
0004
0005 IF #vis AND NOT #edge3 THEN
0006   #Blue[#idx].ID := #idx + 1;
0007   #Blue[#idx].Time_vision := #Actual_Time;
0008   #Blue[#idx].Vision_inspect := TRUE;
0009   #Blue[#idx].Data_Ready := FALSE;
0010   #Blue[#idx].Time_as := DTL#1970-01-01-00:00:00.0;
0011
0012
0013   #Blue[#idx].Time_mach := T_SUB(IN1 := #Actual_Time, IN2 := T#15s);
0014
0015
0016   #Time_deadline := T_ADD(IN1 := #Actual_Time, IN2 := T#30s);
```

Totally Integrated Automation Portal		
<pre> 0017 0018 #idx := #idx + 1; 0019 IF #idx > 50 THEN 0020 #idx := 0; 0021 END_IF; 0022 END_IF; 0023 #edge3 := #vis; 0024 0025 0026 IF #exit AND NOT #edge4 THEN 0027 FOR #j := 0 TO 50 DO 0028 0029 IF #Blue[#j].ID <> 0 AND #Blue[#j].Vision_in- spect AND #Blue[#j].Time_as = DTL#1970-01-01-00:00:00.0 THEN 0030 0031 0032 IF #Actual_Time > T_ADD(IN1 := #Blue[#j].Time_vision, IN2 := T#35s) THEN 0033 #Blue[#j].Time_as := DTL#1990-01-01-00:00:00.0; 0034 CONTINUE; 0035 END_IF; 0036 0037 0038 #Blue[#j].Time_as := #Actual_Time; 0039 0040 0041 IF #Actual_Time <= #Time_deadline THEN 0042 #Blue[#j].Data_Ready := TRUE; 0043 ELSE 0044 #Blue[#j].Data_Ready := FALSE; 0045 END_IF; 0046 0047 EXIT; 0048 END_IF; 0049 END_FOR; 0050 END_IF; 0051 #edge4 := #exit; 0052 0053 // STATS 0054 #nr_conf := 0; 0055 #nr_neconf := 0; 0056 0057 FOR #k := 0 TO 50 DO 0058 IF #Blue[#k].ID <> 0 THEN 0059 0060 IF #Blue[#k].Time_as > DTL#1970-01-01-00:00:00.0 AND #Blue[#k].Time_as <> DTL#1990-01-01-00:00:00.0 THEN 0061 #nr_conf := #nr_conf + 1; 0062 END_IF; 0063 0064 IF #Blue[#k].Time_as = DTL#1990-01-01-00:00:00.0 THEN 0065 #nr_neconf := #nr_neconf + 1; 0066 END_IF; 0067 END_IF; 0068 END_FOR; 0069 0070 0071 #Total_tsum_DINT := 0; 0072 FOR #m := 0 TO 50 DO 0073 </pre>		

Totally Integrated Automation Portal		
<pre>0074 IF #Blue[#m].ID <> 0 AND #Blue[#m].Time_as > DTL#1970-01-01-00:00:00.0 AND #Blue[#m].Time_as <> DTL#1990-01-01-00:00:00.0 THEN 0075 #Diff_time := T_DIFF(IN1 := #Blue[#m].Time_as, IN2 := #Blue[#m].Time_mach); 0076 #Total_tsum_DINT := #Total_tsum_DINT + TIME_TO_DINT(#Diff_time); 0077 END_IF; 0078 END_FOR; 0079 0080 0081 IF #nr_conf > 0 THEN 0082 #av_sec := (DINT_TO_REAL(#Total_tsum_DINT) / INT_TO_RE- AL(#nr_conf)) / 1000.0; 0083 ELSE 0084 #av_sec := 0.0; 0085 END_IF;</pre>		

factoryio [CPU 1511-1 PN] / Program blocks

Blue_DB [DB3]

Blue_DB Properties					
General					
Name	Blue_DB	Number	3	Type	DB
Language	DB	Numbering	Automatic		
Information					
Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Start value	Retain
Input			
Output			
InOut			
▼ Static			
IEC_Timer_0_Instance	TON_TIME		False
IEC_Timer_0_Instance_1	TON_TIME		False
IEC_Timer_0_Instance_2	TON_TIME		False
IEC_Timer_0_Instance_3	TON_TIME		False
IEC_Timer_0_Instance_4	TON_TIME		False
IEC_Timer_0_Instance_5	TON_TIME		False
IEC_Timer_0_Instance_6	TON_TIME		False
IEC_Timer_0_Instance_7	TON_TIME		False
IEC_Timer_0_Instance_8	TON_TIME		False
IEC_Timer_0_Instance_9	TON_TIME		False
IEC_Timer_0_Instance_10	TON_TIME		False
count_blue	CTU_INT		True
timer_pick	TON_TIME		False

Totally Integrated Automation Portal																																																																																						
factoryio [CPU 1511-1 PN] / Program blocks Green_DB [DB5] Green_DB Properties General <table><tr><td>Name</td><td>Green_DB</td><td>Number</td><td>5</td><td>Type</td><td>DB</td></tr><tr><td>Language</td><td>DB</td><td>Numbering</td><td>Automatic</td><td></td><td></td></tr></table> Information <table><tr><td>Title</td><td></td><td>Author</td><td></td><td>Comment</td><td></td></tr><tr><td>Family</td><td></td><td>Version</td><td>0.1</td><td>User-defined ID</td><td></td></tr></table>			Name	Green_DB	Number	5	Type	DB	Language	DB	Numbering	Automatic			Title		Author		Comment		Family		Version	0.1	User-defined ID																																																													
Name	Green_DB	Number	5	Type	DB																																																																																	
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Title		Author		Comment																																																																																		
Family		Version	0.1	User-defined ID																																																																																		
<table><tr><th>Name</th><th>Data type</th><th>Start value</th><th>Retain</th></tr><tr><td>Input</td><td></td><td></td><td></td></tr><tr><td>Output</td><td></td><td></td><td></td></tr><tr><td>InOut</td><td></td><td></td><td></td></tr><tr><td>▼ Static</td><td></td><td></td><td></td></tr><tr><td>IEC_Timer_0_Green</td><td>TON_TIME</td><td></td><td>False</td></tr><tr><td>IEC_Timer_0_Instance_1_Green</td><td>TON_TIME</td><td></td><td>False</td></tr><tr><td>IEC_Timer_0_Instance_2_Green</td><td>TON_TIME</td><td></td><td>False</td></tr><tr><td>IEC_Timer_0_Instance_4_Green</td><td>TON_TIME</td><td></td><td>False</td></tr><tr><td>IEC_Timer_0_Instance_5_Green</td><td>TON_TIME</td><td></td><td>False</td></tr><tr><td>IEC_Timer_0_Instance_6_Green</td><td>TON_TIME</td><td></td><td>False</td></tr><tr><td>IEC_Timer_0_Instance_7_Green</td><td>TON_TIME</td><td></td><td>False</td></tr><tr><td>IEC_Timer_0_Instance_8_Green</td><td>TON_TIME</td><td></td><td>False</td></tr><tr><td>IEC_Timer_0_Instance_10_Green</td><td>TON_TIME</td><td></td><td>False</td></tr><tr><td>Tlmer0_Blue_1</td><td>TON_TIME</td><td></td><td>False</td></tr><tr><td>Tlmer0_Blue_2</td><td>TON_TIME</td><td></td><td>False</td></tr><tr><td>Tlmer0_Blue_3</td><td>TON_TIME</td><td></td><td>False</td></tr><tr><td>Tlmer0_Blue_4</td><td>TON_TIME</td><td></td><td>False</td></tr><tr><td>Tlmer0_Blue_5</td><td>TON_TIME</td><td></td><td>False</td></tr><tr><td>Tlmer0_Blue_6</td><td>TON_TIME</td><td></td><td>False</td></tr><tr><td>Tlmer0_Blue_7</td><td>TON_TIME</td><td></td><td>False</td></tr></table>			Name	Data type	Start value	Retain	Input				Output				InOut				▼ Static				IEC_Timer_0_Green	TON_TIME		False	IEC_Timer_0_Instance_1_Green	TON_TIME		False	IEC_Timer_0_Instance_2_Green	TON_TIME		False	IEC_Timer_0_Instance_4_Green	TON_TIME		False	IEC_Timer_0_Instance_5_Green	TON_TIME		False	IEC_Timer_0_Instance_6_Green	TON_TIME		False	IEC_Timer_0_Instance_7_Green	TON_TIME		False	IEC_Timer_0_Instance_8_Green	TON_TIME		False	IEC_Timer_0_Instance_10_Green	TON_TIME		False	Tlmer0_Blue_1	TON_TIME		False	Tlmer0_Blue_2	TON_TIME		False	Tlmer0_Blue_3	TON_TIME		False	Tlmer0_Blue_4	TON_TIME		False	Tlmer0_Blue_5	TON_TIME		False	Tlmer0_Blue_6	TON_TIME		False	Tlmer0_Blue_7	TON_TIME		False
Name	Data type	Start value	Retain																																																																																			
Input																																																																																						
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IEC_Timer_0_Green	TON_TIME		False																																																																																			
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IEC_Timer_0_Instance_5_Green	TON_TIME		False																																																																																			
IEC_Timer_0_Instance_6_Green	TON_TIME		False																																																																																			
IEC_Timer_0_Instance_7_Green	TON_TIME		False																																																																																			
IEC_Timer_0_Instance_8_Green	TON_TIME		False																																																																																			
IEC_Timer_0_Instance_10_Green	TON_TIME		False																																																																																			
Tlmer0_Blue_1	TON_TIME		False																																																																																			
Tlmer0_Blue_2	TON_TIME		False																																																																																			
Tlmer0_Blue_3	TON_TIME		False																																																																																			
Tlmer0_Blue_4	TON_TIME		False																																																																																			
Tlmer0_Blue_5	TON_TIME		False																																																																																			
Tlmer0_Blue_6	TON_TIME		False																																																																																			
Tlmer0_Blue_7	TON_TIME		False																																																																																			

factoryio [CPU 1511-1 PN] / Program blocks

Load_parts_DB [DB9]

Load_parts_DB Properties

General

Name	Load_parts_DB	Number	9	Type	DB
Language	DB	Numbering	Automatic		

Information

Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Start value	Retain
▼ Input			
stopblp	Bool	false	False
stopgrp	Bool	false	False
boxdet	Bool	false	False
done_robot	Bool	false	False
stopbox	Bool	false	False
▼ Output			
xp	Real	0.0	False
yp	Real	0.0	False
zp	Real	0.0	False
xd	Real	0.0	False
yd	Real	0.0	False
zd	Real	0.0	False
done	Bool	false	False
InOut			
▼ Static			
step	DInt	0	False
counter	Int	0	False
is_box	Bool	false	False
enable_robot	Bool	false	False
inst_Timer	IEC_TIMER		False
delay_active	Bool	false	False
need_blue	Bool	false	False

factoryio [CPU 1511-1 PN] / Program blocks

Move_part_DB [DB11]

Move_part_DB Properties

General

Name	Move_part_DB	Number	11	Type	DB
Language	DB	Numbering	Automatic		

Information

Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Start value	Retain
▼ Input			
enable	Bool	false	False
pxpos	Real	0.0	False
pypos	Real	0.0	False
pzpos	Real	0.0	False
xp	Real	0.0	False
yp	Real	0.0	False
zp	Real	0.0	False
xd	Real	0.0	False
yd	Real	0.0	False
zd	Real	0.0	False
pboxdet	Bool	false	False
detection	Bool	false	False
▼ Output			
pxsetp	Real	0.0	False
pysetp	Real	0.0	False
pzsetp	Real	0.0	False
pgrab	Bool	false	False
done	Bool	false	False
InOut			
▼ Static			
inst_Timer	IEC_TIMER		False
delay_active	Bool	false	False
step	DInt	0	False

factoryio [CPU 1511-1 PN] / Program blocks

Move_part_DB_1 [DB13]

Move_part_DB_1 Properties

General

Name	Move_part_DB_1	Number	13	Type	DB
Language	DB	Numbering	Automatic		

Information

Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Start value	Retain
▼ Input			
enable	Bool	false	False
pxpos	Real	0.0	False
pypos	Real	0.0	False
pzpos	Real	0.0	False
xp	Real	0.0	False
yp	Real	0.0	False
zp	Real	0.0	False
xd	Real	0.0	False
yd	Real	0.0	False
zd	Real	0.0	False
pboxdet	Bool	false	False
detection	Bool	false	False
▼ Output			
pxsetp	Real	0.0	False
pysetp	Real	0.0	False
pzsetp	Real	0.0	False
pgrab	Bool	false	False
done	Bool	false	False
InOut			
▼ Static			
inst_Timer	IEC_TIMER		False
delay_active	Bool	false	False
step	DInt	0	False

factoryio [CPU 1511-1 PN] / Program blocks

Move-blue [DB12]

Move-blue Properties					
General					
Name	Move-blue	Number	12	Type	DB
Language	DB	Numbering	Automatic		
Information					
Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Start value	Retain
▼ Input			
enable	Bool	false	False
pxpos	Real	0.0	False
pypos	Real	0.0	False
pzpos	Real	0.0	False
xp	Real	0.0	False
yp	Real	0.0	False
zp	Real	0.0	False
xd	Real	0.0	False
yd	Real	0.0	False
zd	Real	0.0	False
pboxdet	Bool	false	False
detection	Bool	false	False
▼ Output			
pxsetp	Real	0.0	False
pysetp	Real	0.0	False
pzsetp	Real	0.0	False
pgrab	Bool	false	False
done	Bool	false	False
InOut			
▼ Static			
inst_Timer	IEC_TIMER		False
delay_active	Bool	false	False
step	DInt	0	False

Totally Integrated Automation Portal																																														
factoryio [CPU 1511-1 PN] / Program blocks Pallet+Pick_DB [DB10] Pallet+Pick_DB Properties General <table><tr><td>Name</td><td>Pallet+Pick_DB</td><td>Number</td><td>10</td><td>Type</td><td>DB</td></tr><tr><td>Language</td><td>DB</td><td>Numbering</td><td>Automatic</td><td></td><td></td></tr></table> Information <table><tr><td>Title</td><td></td><td>Author</td><td></td><td>Comment</td><td></td></tr><tr><td>Family</td><td></td><td>Version</td><td>0.1</td><td>User-defined ID</td><td></td></tr></table> <table><tr><th>Name</th><th>Data type</th><th>Start value</th><th>Retain</th></tr><tr><td>Input</td><td></td><td></td><td></td></tr><tr><td>Output</td><td></td><td></td><td></td></tr><tr><td>InOut</td><td></td><td></td><td></td></tr><tr><td>Static</td><td></td><td></td><td></td></tr></table>			Name	Pallet+Pick_DB	Number	10	Type	DB	Language	DB	Numbering	Automatic			Title		Author		Comment		Family		Version	0.1	User-defined ID		Name	Data type	Start value	Retain	Input				Output				InOut				Static			
Name	Pallet+Pick_DB	Number	10	Type	DB																																									
Language	DB	Numbering	Automatic																																											
Title		Author		Comment																																										
Family		Version	0.1	User-defined ID																																										
Name	Data type	Start value	Retain																																											
Input																																														
Output																																														
InOut																																														
Static																																														

factoryio [CPU 1511-1 PN] / Program blocks

pick_pack [DB16]

pick_pack Properties

General

Name	pick_pack	Number	16	Type	DB
Language	DB	Numbering	Automatic		

Information

Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Start value	Retain
▼ Input			
enable	Bool	false	False
pxpos	Real	0.0	False
pypos	Real	0.0	False
pzpos	Real	0.0	False
xp	Real	0.0	False
yp	Real	0.0	False
zp	Real	0.0	False
xd	Real	0.0	False
yd	Real	0.0	False
zd	Real	0.0	False
pboxdet	Bool	false	False
detection	Bool	false	False
▼ Output			
pxsetp	Real	0.0	False
pysetp	Real	0.0	False
pzsetp	Real	0.0	False
pgrab	Bool	false	False
done	Bool	false	False
InOut			
▼ Static			
inst_Timer	IEC_TIMER		False
delay_active	Bool	false	False
step	DInt	0	False

factoryio [CPU 1511-1 PN] / Program blocks

Pick_simplu_1_DB [DB20]

Pick_simplu_1_DB Properties					
General					
Name	Pick_simplu_1_DB	Number	20	Type	DB
Language	DB	Numbering	Automatic		
Information					
Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Start value	Retain
▼ Input			
enable	Bool	false	False
pxpos	Real	0.0	False
pzpos	Real	0.0	False
xp	Real	0.0	False
zp	Real	0.0	False
xd	Real	0.0	False
zd	Real	0.0	False
pboxdet	Bool	false	False
▼ Output			
pxsetp	Real	0.0	False
pzsetp	Real	0.0	False
pgrab	Bool	false	False
done	Bool	false	False
rotccw	Bool	false	False
rotcw	Bool	false	False
InOut			
▼ Static			
inst_Timer	IEC_TIMER		False
delay_active	Bool	false	False
old_step	DInt	0	False
step	DInt	0	False

factoryio [CPU 1511-1 PN] / Program blocks

Pick_simplu_DB [DB6]

Pick_simplu_DB Properties					
General					
Name	Pick_simplu_DB	Number	6	Type	DB
Language	DB	Numbering	Automatic		
Information					
Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Start value	Retain
▼ Input			
enable	Bool	false	False
pxpos	Real	0.0	False
pzpos	Real	0.0	False
xp	Real	0.0	False
zp	Real	0.0	False
xd	Real	0.0	False
zd	Real	0.0	False
pboxdet	Bool	false	False
▼ Output			
pxsetp	Real	0.0	False
pzsetp	Real	0.0	False
pgrab	Bool	false	False
done	Bool	false	False
rotccw	Bool	false	False
rotcw	Bool	false	False
InOut			
▼ Static			
inst_Timer	IEC_TIMER		False
delay_active	Bool	false	False
old_step	DInt	0	False
step	DInt	0	False

Totally Integrated Automation Portal		
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factoryio [CPU 1511-1 PN] / Program blocks

ctf_gr [DB14]

ctf_gr Properties

General

Name	ctf_gr	Number	14	Type	DB
Language	DB	Numbering	Automatic		

Information

Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Start value	Retain
▼ Input			
sensor	Bool	false	False
reset	Bool	false	False
PV	Int	0	False
▼ Output			
done	Bool	false	False
InOut			
▼ Static			
CV	Int	0	False
sensor_prev	Bool	false	False

factoryio [CPU 1511-1 PN] / Program blocks

Packaging_pick [DB17]

Packaging_pick Properties

General

Name	Packaging_pick	Number	17	Type	DB
Language	DB	Numbering	Automatic		

Information

Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Start value	Retain
▼ Input			
enable	Bool	false	False
pxpos	Real	0.0	False
pzpos	Real	0.0	False
xp	Real	0.0	False
zp	Real	0.0	False
xd	Real	0.0	False
zd	Real	0.0	False
pboxdet	Bool	false	False
▼ Output			
pxsetp	Real	0.0	False
pzsetp	Real	0.0	False
pgrab	Bool	false	False
done	Bool	false	False
rotccw	Bool	false	False
rotcw	Bool	false	False
InOut			
▼ Static			
inst_Timer	IEC_TIMER		False
delay_active	Bool	false	False
old_step	DInt	0	False
step	DInt	0	False

factoryio [CPU 1511-1 PN] / Program blocks

Trasabilitate_blue_DB [DB18]

Trasabilitate_blue_DB Properties					
General					
Name	Trasabilitate_blue_DB	Number	18	Type	DB
Language	DB	Numbering	Automatic		
Information					
Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Start value	Retain
▼ Input			
vis	Bool	false	False
exit	Bool	false	False
Output			
InOut			
▼ Static			
Blue	Array[0..50] of "udt_Blue_Trace"		False
idx	Int	0	False
edge3	Bool	false	False
edge4	Bool	false	False
Time_status	Int	0	False
Time_deadline	DTL	DTL#1970-01-01-00:00:00	False
Actual_Time	DTL	DTL#1970-01-01-00:00:00	False
j	Int	0	False
k	Int	0	False
nr_conf	Int	0	False
nr_neconf	Int	0	False
Total_tsum_DINT	DInt	0	False
Diff_time	Time	T#0ms	False
av_sec	Real	0.0	False
m	Int	0	False

factoryio [CPU 1511-1 PN] / Program blocks

Trasabilitate_green [FB10]

Trasabilitate_green Properties					
General					
Name	Trasabilitate_green	Number	10	Type	FB
Language	SCL	Numbering	Automatic		
Information					
Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Default value
▼ Input		
vis	Bool	false
stop	Bool	false
Output		
InOut		
▼ Static		
Green	Array[0..50] of "udt_Green_Trace"	
idx	Int	0
Exit_Delay	IEC_TIMER	
edge3	Bool	false
edge4	Bool	false
Actual_Time	DTL	DTL#1970-01-01-00:00:00
j	Int	0
Time_deadline	DTL	DTL#1970-01-01-00:00:00
Time_status	Int	0
nr_conf	Int	0
k	Int	0
nr_neconf	Int	0
m	Int	0
Diff_time	Time	T#0ms
av_sec	Real	0.0
Total_tsum_DINT	DInt	0
Temp		
Constant		

```
0001 #Time_status := RD_LOC_T(OUT => #Actual_Time);
0002
0003
0004 IF #vis AND NOT #edge3 THEN
0005     #Green[#idx].ID := #idx + 1;
0006     #Green[#idx].Time_vision := #Actual_Time;
0007     #Green[#idx].Vision_inspect := TRUE;
0008     #Green[#idx].Data_Ready := FALSE;
0009     #Green[#idx].Time_as := DTL#1970-01-01-00:00:00.0;
0010
0011     #Green[#idx].Time_mach := T_SUB(IN1 := #Actual_Time, IN2 := T#15s);
0012
0013
0014     #Time_deadline := T_ADD(IN1 := #Actual_Time, IN2 := T#30s);
0015
```

Totally Integrated Automation Portal		
<pre> 0016 #idx := #idx + 1; 0017 IF #idx > 50 THEN 0018 #idx := 0; 0019 END_IF; 0020 END_IF; 0021 #edge3 := #vis; 0022 0023 IF #stop AND NOT #edge4 THEN 0024 FOR #j := 0 TO 50 DO 0025 0026 IF #Green[#j].ID <> 0 AND #Green[#j].Vision_in- spect AND #Green[#j].Time_as = DTL#1970-01-01-00:00:00.0 THEN 0027 0028 IF #Actual_Time > T_ADD(IN1 := #Green[#j].Time_vi- sion, IN2 := T#35s) THEN 0029 #Green[#j].Time_as := DTL#1990-01-01-00:00:00.0; 0030 CONTINUE; 0031 END_IF; 0032 0033 #Green[#j].Time_as := #Actual_Time; 0034 0035 IF #Actual_Time <= #Time_deadline THEN 0036 #Green[#j].Data_Ready := TRUE; 0037 ELSE 0038 #Green[#j].Data_Ready := FALSE; 0039 END_IF; 0040 0041 EXIT; 0042 END_IF; 0043 END_FOR; 0044 END_IF; 0045 #edge4 := #stop; 0046 0047 // hmi stats 0048 #nr_conf := 0; 0049 #nr_neconf := 0; 0050 FOR #k := 0 TO 50 DO 0051 IF #Green[#k].ID <> 0 THEN 0052 0053 IF #Green[#k].Time_as > DTL#1970-01-01-00:00:00.0 0054 AND #Green[#k].Time_as <> DTL#1990-01-01-00:00:00.0 THEN 0055 #nr_conf := #nr_conf + 1; 0056 END_IF; 0057 0058 IF #Green[#k].Time_as = DTL#1990-01-01-00:00:00.0 THEN 0059 #nr_neconf := #nr_neconf + 1; 0060 END_IF; 0061 END_IF; 0062 END_FOR; 0063 0064 // average time mach->asm 0065 #Total_tsum_DINT := 0; 0066 FOR #m := 0 TO 50 DO 0067 0068 IF #Green[#m].ID <> 0 AND #Green[#m].Time_as > DTL#1970-01-01-00:00:00.0 AND #Green[#m].Time_as <> DTL#1990-01-01-00:00:00.0 THEN 0069 #Diff_time := T_DIFF(IN1 := #Green[#m].Time_as, IN2 := #Green[#m].Time_mach); 0070 </pre>		

Totally Integrated Automation Portal		
<pre>0071 #Total_tsum_DINT := #Total_tsum_DINT + TIME_TO_DINT(#Diff_time); 0072 END_IF; 0073 END_FOR; 0074 0075 0076 IF #nr_conf > 0 THEN 0077 #av_sec := (DINT_TO_REAL(#Total_tsum_DINT) / INT_TO_REAL(#nr_conf)) / 1000.0; 0078 ELSE 0079 #av_sec := 0.0; 0080 END_IF;</pre>		

factoryio [CPU 1511-1 PN] / Program blocks

Trasabilitate_green_DB [DB2]

Trasabilitate_green_DB Properties

General

Name	Trasabilitate_green_DB	Number	2	Type	DB
Language	DB	Numbering	Automatic		

Information

Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Start value	Retain
▼ Input			
vis	Bool	false	False
stop	Bool	false	False
Output			
InOut			
▼ Static			
Green	Array[0..50] of "udt_Green_Trace"		False
idx	Int	0	False
Exit_Delay	IEC_TIMER		False
edge3	Bool	false	False
edge4	Bool	false	False
Actual_Time	DTL	DTL#1970-01-01-00:00:00	False
j	Int	0	False
Time_deadline	DTL	DTL#1970-01-01-00:00:00	False
Time_status	Int	0	False
nr_conf	Int	0	False
k	Int	0	False
nr_neconf	Int	0	False
m	Int	0	False
Diff_time	Time	T#0ms	False
av_sec	Real	0.0	False
Total_tsum_DINT	DInt	0	False

factoryio [CPU 1511-1 PN] / Program blocks

Trasabilitate_box [FB12]

Trasabilitate_box Properties

General

Name	Trasabilitate_box	Number	12	Type	FB
Language	SCL	Numbering	Automatic		

Information

Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Default value
▼ Input		
stopbox	Bool	false
stopbox2	Bool	false
doneLoader	Bool	false
donePacker	Bool	false
boxdet	Bool	false
Output		
InOut		
▼ Static		
Box	Array[0..25] of "udt_Box_Trace"	
idxBox	Int	0
Exit_Delay	Bool	false
edgeBox1	Bool	false
edgeBox2	Bool	false
edgeDonePack	Bool	false
edgeDoneLoad	Bool	false
Actual_Time	DTL	DTL#1970-01-01-00:00:00
k	Int	0
Time_status	Int	0
marker	Bool	false
nr_inc	Int	0
nr_com	Int	0
Total_System_Time	Time	T#0ms
Avg_System_Time	Real	0.0
Avg_Wait_Asm	Real	0.0
Total_Wait_Asm	Time	T#0ms
Count_Finished	Int	0
j	Int	0
▼ Temp		
Temp_Time_Diff	Time	
t_ms	DInt	
r_ms	Real	
r_count	Real	
Constant		

```
0001 #Time_status := RD_LOC_T(OUT => #Actual_Time);
0002
0003
0004 IF #idxBox > 24 OR #idxBox < 0 THEN
```

Totally Integrated Automation Portal		
<pre> 0005 #idxBox := 0; 0006 END_IF; 0007 0008 0009 IF #boxdet THEN 0010 #marker := TRUE; 0011 END_IF; 0012 0013 0014 IF #stopbox AND NOT #edgeBox1 THEN 0015 #Box[#idxBox].ID := #idxBox + 1; 0016 #Box[#idxBox].Time_asm := #Actual_Time; 0017 #Box[#idxBox].Box/Pallet := #marker; 0018 #marker := FALSE; 0019 0020 0021 #Box[#idxBox].Time_asdon := DTL#1970-01-01-00:00:00.0; 0022 #Box[#idxBox].Time_pack := DTL#1970-01-01-00:00:00.0; 0023 #Box[#idxBox].Time_exsys := DTL#1970-01-01-00:00:00.0; 0024 #Box[#idxBox].Assembly := FALSE; 0025 #Box[#idxBox].Packaging := FALSE; 0026 #Box[#idxBox].data_Ready := FALSE; 0027 0028 #idxBox := #idxBox + 1; 0029 IF #idxBox > 24 THEN 0030 #idxBox := 0; 0031 END_IF; 0032 END_IF; 0033 #edgeBox1 := #stopbox; 0034 0035 0036 IF #doneLoader AND NOT #edgeDoneLoad THEN 0037 FOR #k := 0 TO 50 DO 0038 IF #Box[#k].ID <> 0 AND NOT #Box[#k].Assembly THEN 0039 #Box[#k].Time_asdon := #Actual_Time; 0040 #Box[#k].Assembly := TRUE; 0041 0042 #Temp_Time_Diff := T_DIFF(IN1 := #Box[#k].Time_as- don, IN2 := #Box[#k].Time_asm); 0043 #Total_Wait_Asm := #Total_Wait_Asm + #Temp_Time_Diff; 0044 0045 EXIT; 0046 END_IF; 0047 END_FOR; 0048 END_IF; 0049 #edgeDoneLoad := #doneLoader; 0050 0051 IF #stopbox2 AND NOT #edgeBox2 THEN 0052 FOR #k := 0 TO 50 DO 0053 IF #Box[#k].Assem- bly AND #Box[#k].Time_pack = DTL#1970-01-01-00:00:00.0 THEN 0054 #Box[#k].Time_pack := #Actual_Time; 0055 EXIT; 0056 END_IF; 0057 END_FOR; 0058 END_IF; 0059 #edgeBox2 := #stopbox2; 0060 0061 IF #donePacker AND NOT #edgeDonePack THEN </pre>		

Totally Integrated Automation Portal		
<pre> 0062 FOR #k := 0 TO 50 DO 0063 IF #Box[#k].Time_pack <> DTL#1970-01-01-00:00:00.0 AND NOT #Box[#k].Pack- aging THEN 0064 #Box[#k].Time_exsys := #Actual_Time; 0065 #Box[#k].Packaging := TRUE; 0066 #Box[#k].data_Ready := TRUE; 0067 #Temp_Time_Diff := T_DIFF(IN1 := #Box[#k].Time_ex- sys, IN2 := #Box[#k].Time_asm); 0068 #Total_System_Time := #Total_System_Time + #Temp_Time_Diff; 0069 #Count_Finished := #Count_Finished + 1; 0070 0071 EXIT; 0072 END_IF; 0073 END_FOR; 0074 END_IF; 0075 #edgeDonePack := #donePacker; 0076 0077 0078 IF #Count_Finished > 0 THEN 0079 0080 #t_ms := TIME_TO_DINT(#Total_Wait_Asm); 0081 #r_ms := #t_ms; 0082 #r_count := #Count_Finished; 0083 #Avg_Wait_Asm := (#r_ms / #r_count) / 1000.0; 0084 #t_ms := TIME_TO_DINT(#Total_System_Time); 0085 #r_ms := #t_ms; 0086 #Avg_System_Time := (#r_ms / #r_count) / 1000.0; 0087 0088 END_IF; 0089 0090 #nr_com := 0; 0091 #nr_inc := 0; 0092 0093 FOR #j := 0 TO 24 DO 0094 IF #Box[#j].ID <> 0 THEN 0095 IF #Box[#j].Time_exsys <> DTL#1970-01-01-00:00:00.0 AND #Box[#j].Packag- ing THEN 0096 #nr_com := #nr_com + 1; 0097 ELSE 0098 #nr_inc := #nr_inc + 1; 0099 END_IF; 0100 END_IF; 0101 END_FOR; </pre>		

factoryio [CPU 1511-1 PN] / Program blocks

Trasabilitate_box_DB [DB8]

Trasabilitate_box_DB Properties

General

Name	Trasabilitate_box_DB	Number	8	Type	DB
Language	DB	Numbering	Automatic		

Information

Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Start value	Retain
▼ Input			
stopbox	Bool	false	False
stopbox2	Bool	false	False
doneLoader	Bool	false	False
donePacker	Bool	false	False
boxdet	Bool	false	False
Output			
InOut			
▼ Static			
Box	Array[0..25] of "udt_Box_Trace"		False
idxBox	Int	0	False
Exit_Delay	Bool	false	False
edgeBox1	Bool	false	False
edgeBox2	Bool	false	False
edgeDonePack	Bool	false	False
edgeDoneLoad	Bool	false	False
Actual_Time	DTL	DTL#1970-01-01-00:00:00	False
k	Int	0	False
Time_status	Int	0	False
marker	Bool	false	False
nr_inc	Int	0	False
nr_com	Int	0	False
Total_System_Time	Time	T#0ms	False
Avg_System_Time	Real	0.0	False
Avg_Wait_Asm	Real	0.0	False
Total_Wait_Asm	Time	T#0ms	False
Count_Finished	Int	0	False
j	Int	0	False

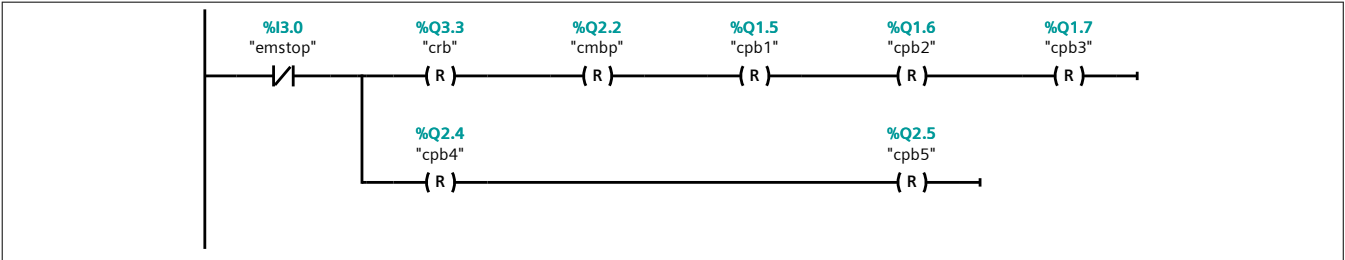
factoryio [CPU 1511-1 PN] / Program blocks

em_stop [FB13]

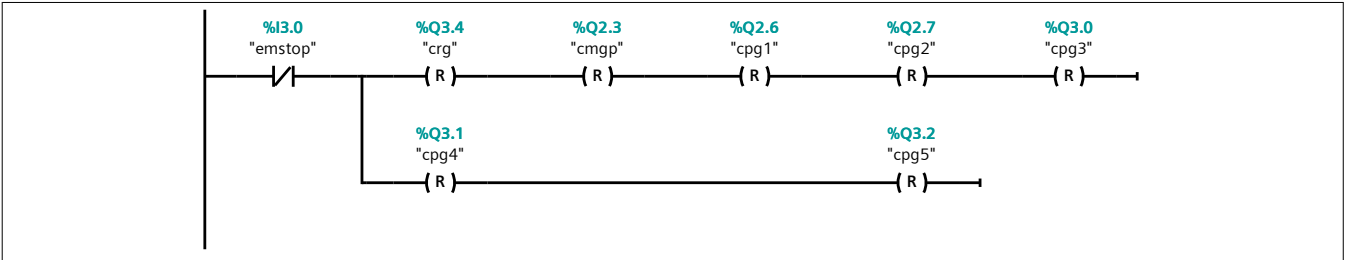
em_stop Properties					
General					
Name	em_stop	Number	13	Type	FB
Language	LAD	Numbering	Automatic		
Information					
Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Default value	Retain
Input			
Output			
InOut			
Static			
Temp			
Constant			

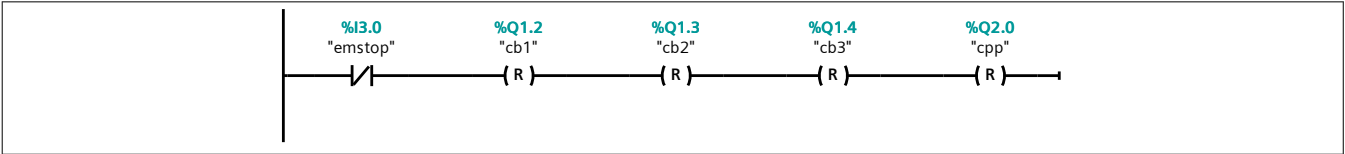
Network 1:



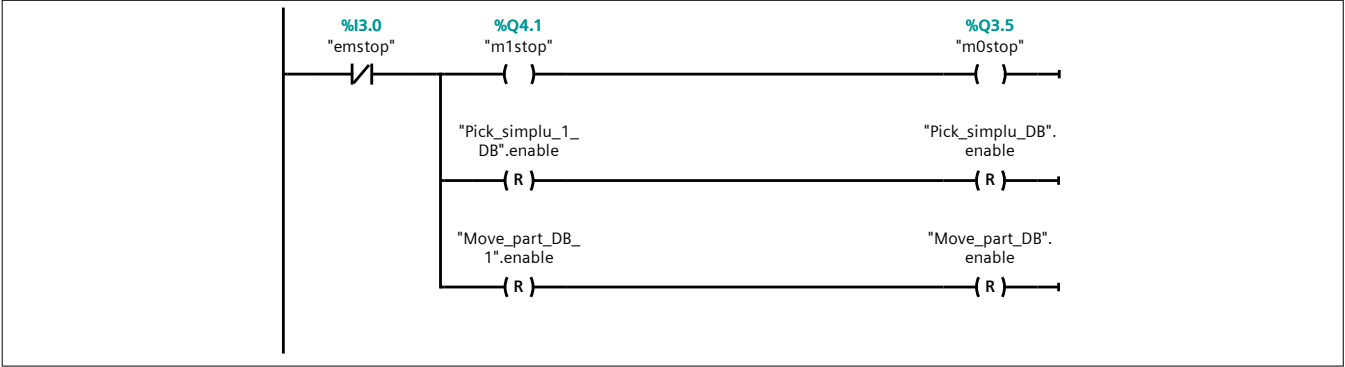
Network 2:



Network 3:



Network 4:



Totally Integrated Automation Portal																																														
factoryio [CPU 1511-1 PN] / Program blocks em_stop_DB [DB19] em_stop_DB Properties General <table><tr><td>Name</td><td>em_stop_DB</td><td>Number</td><td>19</td><td>Type</td><td>DB</td></tr><tr><td>Language</td><td>DB</td><td>Numbering</td><td>Automatic</td><td></td><td></td></tr></table> Information <table><tr><td>Title</td><td></td><td>Author</td><td></td><td>Comment</td><td></td></tr><tr><td>Family</td><td></td><td>Version</td><td>0.1</td><td>User-defined ID</td><td></td></tr></table> <table><tr><th>Name</th><th>Data type</th><th>Start value</th><th>Retain</th></tr><tr><td>Input</td><td></td><td></td><td></td></tr><tr><td>Output</td><td></td><td></td><td></td></tr><tr><td>InOut</td><td></td><td></td><td></td></tr><tr><td>Static</td><td></td><td></td><td></td></tr></table>			Name	em_stop_DB	Number	19	Type	DB	Language	DB	Numbering	Automatic			Title		Author		Comment		Family		Version	0.1	User-defined ID		Name	Data type	Start value	Retain	Input				Output				InOut				Static			
Name	em_stop_DB	Number	19	Type	DB																																									
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Totally Integrated Automation Portal																																																																						
factoryio [CPU 1511-1 PN] / Program blocks bluecount [DB21] bluecount Properties General <table><tr><td>Name</td><td>bluecount</td><td>Number</td><td>21</td><td>Type</td><td>DB</td></tr><tr><td>Language</td><td>DB</td><td>Numbering</td><td>Automatic</td><td></td><td></td></tr></table> Information <table><tr><td>Title</td><td></td><td>Author</td><td></td><td>Comment</td><td></td></tr><tr><td>Family</td><td></td><td>Version</td><td>0.1</td><td>User-defined ID</td><td></td></tr></table> <table><thead><tr><th>Name</th><th>Data type</th><th>Start value</th><th>Retain</th></tr></thead><tbody><tr><td>▼ Input</td><td></td><td></td><td></td></tr><tr><td>sensor</td><td>Bool</td><td>false</td><td>False</td></tr><tr><td>reset</td><td>Bool</td><td>false</td><td>False</td></tr><tr><td>PV</td><td>Int</td><td>0</td><td>False</td></tr><tr><td>▼ Output</td><td></td><td></td><td></td></tr><tr><td>done</td><td>Bool</td><td>false</td><td>False</td></tr><tr><td>InOut</td><td></td><td></td><td></td></tr><tr><td>▼ Static</td><td></td><td></td><td></td></tr><tr><td>CV</td><td>Int</td><td>0</td><td>False</td></tr><tr><td>sensor_prev</td><td>Bool</td><td>false</td><td>False</td></tr></tbody></table>			Name	bluecount	Number	21	Type	DB	Language	DB	Numbering	Automatic			Title		Author		Comment		Family		Version	0.1	User-defined ID		Name	Data type	Start value	Retain	▼ Input				sensor	Bool	false	False	reset	Bool	false	False	PV	Int	0	False	▼ Output				done	Bool	false	False	InOut				▼ Static				CV	Int	0	False	sensor_prev	Bool	false	False
Name	bluecount	Number	21	Type	DB																																																																	
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reset	Bool	false	False																																																																			
PV	Int	0	False																																																																			
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done	Bool	false	False																																																																			
InOut																																																																						
▼ Static																																																																						
CV	Int	0	False																																																																			
sensor_prev	Bool	false	False																																																																			

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factoryio [CPU 1511-1 PN] / Program blocks

Startup [OB100]

Startup Properties

General

Name	Startup	Number	100	Type	OB
Language	SCL	Numbering	Automatic		

Information

Title	"Complete Restart"	Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Default value
▼ Input		
LostRetentive	Bool	
LostRTC	Bool	
Temp		
Constant		

0001 IF NOT "Factory I/O (Run)" THEN

0002 "a3apiesa" := TRUE; // flag - Q5.0 a cazut

0003 END_IF;

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factoryio [CPU 1511-1 PN] / Program blocks / System blocks / Program resources

IEC_Timer_0_DB [DB1]

IEC_Timer_0_DB Properties

General

Name	IEC_Timer_0_DB	Number	1	Type	DB
Language	DB	Numbering	Automatic		

Information

Title		Author	Simatic	Comment	
Family	IEC	Version	1.0	User-defined ID	IEC_TMR

Name	Data type	Start value	Retain
▼ Static			
PT	Time	T#0ms	False
ET	Time	T#0ms	False
IN	Bool	false	False
Q	Bool	false	False

Totally Integrated Automation Portal

factoryio [CPU 1511-1 PN] / Program blocks / System blocks / Program resources

timer_packg [DB4]

timer_packg Properties

General

Name	timer_packg	Number	4	Type	DB
Language	DB	Numbering	Automatic		

Information

Title		Author	Simatic	Comment	
Family	IEC	Version	1.0	User-defined ID	IEC_TMR

Name	Data type	Start value	Retain
▼ Static			
PT	Time	T#0ms	False
ET	Time	T#0ms	False
IN	Bool	false	False
Q	Bool	false	False

Totally Integrated Automation Portal

factoryio [CPU 1511-1 PN] / Program blocks / System blocks / Program resources

timer_Evac [DB15]

timer_Evac Properties

General

Name	timer_Evac	Number	15	Type	DB
Language	DB	Numbering	Automatic		

Information

Title		Author	Simatic	Comment	
Family	IEC	Version	1.0	User-defined ID	IEC_TMR

Name	Data type	Start value	Retain
▼ Static			
PT	Time	T#0ms	False
ET	Time	T#0ms	False
IN	Bool	false	False
Q	Bool	false	False

factoryio [CPU 1511-1 PN]

Technology objects

This folder is empty.

factoryio [CPU 1511-1 PN] / PLC tags / Standard-Variablentabelle [52]

PLC tags

PLC tags				
	Name	Data type	Address	Retain
	Tag_1	Bool	%M200.0	False

factoryio [CPU 1511-1 PN] / PLC tags / Standard-Variablentabelle [52]

User constants

User constants		
Name	Data type	Value

factoryio [CPU 1511-1 PN] / PLC tags / inputs_factoryio [33]

PLC tags

PLC tags				
	Name	Data type	Address	Retain
	visgr	Bool	%I1.6	False
	visbl	Bool	%I1.7	False
	boxdet	Bool	%I2.0	False
	stopbl	Bool	%I2.1	False
	stopblp	Bool	%I2.2	False
	stopbox	Bool	%I2.3	False
	stopbox2	Bool	%I2.4	False
	stopgr	Bool	%I2.5	False
	stopgrp	Bool	%I2.6	False
	m0err	Bool	%I3.1	False
	m0open	Bool	%I3.2	False
	m0busy	Bool	%I3.3	False
	m1err	Bool	%I3.5	False
	m1op	Bool	%I3.6	False
	m1busy	Bool	%I3.7	False
	p0boxdet	Bool	%I4.1	False
	p1boxdet	Bool	%I4.2	False
	p2boxdet	Bool	%I4.3	False
	p3boxdet	Bool	%I4.4	False
	m0prog	DInt	%ID4	False
	m1prog	DInt	%ID8	False
	p0xpos	Real	%ID12	False
	p0zpos	Real	%ID16	False
	p1xpos	Real	%ID20	False
	p1zpos	Real	%ID24	False
	p2xpos	Real	%ID28	False
	p2ypos	Real	%ID32	False
	p2zpos	Real	%ID36	False
	p3xpos	Real	%ID40	False
	p3ypos	Real	%ID44	False
	p3zpos	Real	%ID48	False
	stoppack	Bool	%I2.7	False
	emstop	Bool	%I3.0	False

factoryio [CPU 1511-1 PN] / PLC tags / inputs_factoryio [33]

User constants

User constants		
Name	Data type	Value

factoryio [CPU 1511-1 PN] / PLC tags / outputs_factoryio [45]

PLC tags

PLC tags				
	Name	Data type	Address	Retain
	cb1	Bool	%Q1.2	False
	cb2	Bool	%Q1.3	False
	cb3	Bool	%Q1.4	False
	cpb1	Bool	%Q1.5	False
	cpb2	Bool	%Q1.6	False
	cpb3	Bool	%Q1.7	False
	cpp	Bool	%Q2.0	False
	cmbp	Bool	%Q2.2	False
	cmgp	Bool	%Q2.3	False
	cpb4	Bool	%Q2.4	False
	cpb5	Bool	%Q2.5	False
	cpg1	Bool	%Q2.6	False
	cpg2	Bool	%Q2.7	False
	cpg3	Bool	%Q3.0	False
	cpg4	Bool	%Q3.1	False
	cpg5	Bool	%Q3.2	False
	crb	Bool	%Q3.3	False
	crg	Bool	%Q3.4	False
	m0stop	Bool	%Q3.5	False
	m0reset	Bool	%Q3.6	False
	m0start	Bool	%Q3.7	False
	m0prodlids	Bool	%Q4.0	False
	m1stop	Bool	%Q4.1	False
	m1reset	Bool	%Q4.2	False
	m1start	Bool	%Q4.3	False
	m1prodlids	Bool	%Q4.4	False
	roccw	Bool	%Q4.5	False
	rotcw	Bool	%Q4.7	False
	p0grab	Bool	%Q5.1	False
	p1grab	Bool	%Q5.2	False
	p2grab	Bool	%Q5.3	False
	p3grab	Bool	%Q5.4	False
	p0xsetp	Real	%QD4	False
	p0zsetp	Real	%QD8	False
	p1xsetp	Real	%QD12	False
	p1zsetp	Real	%QD16	False
	p2xsetp	Real	%QD20	False

factoryio [CPU 1511-1 PN] / PLC tags / outputs_factoryio [45]

User constants

User constants		
Name	Data type	Value

factoryio [CPU 1511-1 PN] / PLC tags / Standard-Variablentabelle_1 [48]

PLC tags

PLC tags				
	Name	Data type	Address	Retain
	a3apiesa	Bool	%M102.5	False
	conveyor_global	Bool	%M100.1	False
	a2piesa	Bool	%M11.1	False
	resetContG	Bool	%M12.0	False
	resetContB	Bool	%M100.3	False
	BOX	Bool	%M100.4	False
	doneG	Bool	%M100.5	False
	doneB	Bool	%M100.6	False
	setPack	Bool	%M100.2	False
	doneP3	Bool	%M100.7	False
	doneLoader	Bool	%M101.0	False
	Tag_5	Bool	%M102.0	False
	palletpickM	Bool	%M102.1	False
	reset_contor_bl	Bool	%M102.2	False
	BOX2	Bool	%M102.3	False
	donePack	Bool	%M102.4	False
	Tag_6	Bool	%M200.2	False
	Tag_7	Bool	%M200.3	False
	cntBlue	Bool	%M200.4	False

factoryio [CPU 1511-1 PN] / PLC tags / Standard-Variablentabelle_1 [48]

User constants

User constants			
	Name	Data type	Value
	Keiner(1)	Pip	65535
	Automatische Aktualisierung(1)	Pip	0
	TPA 1(1)	Pip	1
	TPA 2(1)	Pip	2
	TPA 3(1)	Pip	3
	TPA 4(1)	Pip	4
	TPA OB Servo(1)	Pip	32768
	Local~MC(1)	Hw_SubModule	51
	Local~Exec(1)	Hw_SubModule	52
	Local(1)	Hw_SubModule	49
	Local~PROFINET-Schnittstelle_1(1)	Hw_Interface	64
	Local~PROFINET-Schnittstelle_1~Port_1(1)	Hw_Interface	65
	Local~HSC_1	Hw_Hsc	257
	Local~HSC_2	Hw_Hsc	258
	Local~HSC_3	Hw_Hsc	259
	Local~HSC_4	Hw_Hsc	260
	Local~HSC_5	Hw_Hsc	261
	Local~HSC_6	Hw_Hsc	262
	Local~AI_2_1	Hw_SubModule	263
	Local~DI_6_DQ_4_1	Hw_SubModule	264
	Local~Pulse_1	Hw_Pwm	265
	Local~Pulse_2	Hw_Pwm	266
	Local~Pulse_3	Hw_Pwm	267
	Local~Pulse_4	Hw_Pwm	268
	OB_Main(1)	OB_PCYLE	1
	Local~Common(1)	Hw_SubModule	50
	Local~Device(1)	Hw_Device	32
	Local~Configuration(1)	Hw_SubModule	33
	Local~OPC-UA	Hw_SubModule	117

factoryio [CPU 1511-1 PN] / PLC tags / Tag table_1 [37]

PLC tags

PLC tags				
	Name	Data type	Address	Retain
	I_High_sensor	Bool	%I0.0	False
	I_Low_sensor	Bool	%I0.1	False
	I_Pallet_sensor	Bool	%I0.2	False
	I_Loaded	Bool	%I0.3	False
	I_At_left_entry	Bool	%I0.4	False
	I_At_left_exit	Bool	%I0.5	False
	I_At_right_entry	Bool	%I0.6	False
	I_At_right_exit	Bool	%I0.7	False
	O_Conveyor_entry	Bool	%Q0.0	False
	O_Load	Bool	%Q0.1	False
	O_Unload	Bool	%Q0.2	False
	O_Transf_left	Bool	%Q0.3	False
	O_Tranf_right	Bool	%Q0.4	False
	O_Conveyor_left	Bool	%Q0.5	False
	Loading	Bool	%M0.0	False
	LoaderBusy	Bool	%M0.1	False
	TransferringRight	Bool	%M0.2	False
	TransferringLeft	Bool	%M0.3	False
	HighBox	Bool	%M0.4	False
	I_Start	Bool	%I1.0	False
	I_Reset	Bool	%I1.1	False
	I_Stop	Bool	%I1.2	False
	I_Emergency	Bool	%I1.3	False
	I_Auto	Bool	%I1.4	False
	I_Factory_Running	Bool	%I1.5	False
	O_Conveyor_right	Bool	%Q0.6	False
	O_Start_light	Bool	%Q0.7	False
	O_Reset_light	Bool	%Q1.0	False
	O_Stop_light	Bool	%Q1.1	False
	P_High_Sensor	Bool	%M1.0	False
	P>Loading	Bool	%M1.1	False
	P_At_left_entry	Bool	%M1.4	False
	P_At_left_exit	Bool	%M1.5	False
	P_At_right_entry	Bool	%M1.6	False
	P_At_right_exit	Bool	%M1.7	False
	P_Factory_Running	Bool	%M2.5	False

factoryio [CPU 1511-1 PN] / PLC tags / Tag table_1 [37]

User constants

User constants		
Name	Data type	Value

factoryio [CPU 1511-1 PN] / PLC data types

udt_Blue_Trace

udt_Blue_Trace Properties					
General					
Name	udt_Blue_Trace	Number	1	Type	UDT
Language		Numbering			
Information					
Title		Author		Comment	
Family		Version		User-defined ID	

Name	Data type	Default value
ID	DInt	0
Time_mach	DTL	DTL#1970-01-01-00:00:00
Vision_inspect	Bool	false
Time_vision	DTL	DTL#1970-01-01-00:00:00
Time_as	DTL	DTL#1970-01-01-00:00:00
Data_Ready	Bool	false

factoryio [CPU 1511-1 PN] / PLC data types

udt_Green_Trace

udt_Green_Trace Properties					
General					
Name	udt_Green_Trace	Number	2	Type	UDT
Language		Numbering			
Information					
Title		Author		Comment	
Family		Version		User-defined ID	

Name	Data type	Default value
ID	DInt	0
Time_mach	DTL	DTL#1970-01-01-00:00:00
Vision_inspect	Bool	false
Time_vision	DTL	DTL#1970-01-01-00:00:00
Time_as	DTL	DTL#1970-01-01-00:00:00
Data_Ready	Bool	false

factoryio [CPU 1511-1 PN] / PLC data types

udt_Box_Trace

udt_Box_Trace Properties					
General					
Name	udt_Box_Trace	Number	3	Type	UDT
Language		Numbering			
Information					
Title		Author		Comment	
Family		Version		User-defined ID	

Name	Data type	Default value
ID	Int	0
Box/Pallet	Bool	false
Time_asm	DTL	DTL#1970-01-01-00:00:00
Time_asdon	DTL	DTL#1970-01-01-00:00:00
Assembly	Bool	false
Time_pack	DTL	DTL#1970-01-01-00:00:00
Time_exsys	DTL	DTL#1970-01-01-00:00:00
Packaging	Bool	false
data_Ready	Bool	false

factoryio [CPU 1511-1 PN] / PLC data types

System data types

This folder is empty.

factoryio [CPU 1511-1 PN] / Watch and force tables

Forcetabelle

Name	Address	Display format	Force value
"rfid0comm":P	%QD19:P	DEC+/-	
"rfid0write":P	%QD51:P	DEC+/-	
"rfid0idx":P	%QD7:P	DEC+/-	
"rfid0exec":P	%Q5.0:P	Bool	

Totally Integrated Automation Portal			
factoryio [CPU 1511-1 PN] / Watch and force tables			
Watch table_1			
Name	Address	Display format	Modify value
"Factory I/O (Run)"	%Q5.0	Bool	
"crb"	%Q3.3	Bool	
"m0reset"	%Q3.6	Bool	
"cmbp"	%Q2.2	Bool	
"m0start"	%Q3.7	Bool	
"conveyor_global"	%M100.1	Bool	
"resetContB"	%M100.3	Bool	
"a3apiesa"	%M102.5	Bool	
"doneB"	%M100.6	Bool	
"cntBlue"	%M200.4	Bool	

factoryio [CPU 1511-1 PN]

Traces

Name

Totally Integrated Automation Portal		
factoryio [CPU 1511-1 PN] / Traces Measurements This folder is empty.		

factoryio [CPU 1511-1 PN] / Traces

Combined measurements

Name

factoryio [CPU 1511-1 PN] / PLC supervisions & alarms

Supervisions

This folder is empty.

factoryio [CPU 1511-1 PN] / PLC supervisions & alarms

PLC alarms

PLC alarms					
Name	Type	ID	Alarm text	Info text	Information only

factoryio [CPU 1511-1 PN] / PLC supervisions & alarms

System alarms

System alarms	
Name	SDIAG_ALCAT_CPU_INFO_MSG_000F
Type	PLC alarm
ID	1
Alarm text	CPU info: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_CPU_ERR_MSG_0010
Type	PLC alarm
ID	2
Alarm text	CPU error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_CPU_ERR_MSG_0110
Type	PLC alarm
ID	3
Alarm text	CPU error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_CPU_MD_MSG_0011
Type	PLC alarm
ID	4
Alarm text	CPU maintenance demanded: @1W%t#7W@ @6W%t#257K@ / @5W%t#7W@ @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_CPU_MD_MSG_0111
Type	PLC alarm
ID	5
Alarm text	CPU maintenance demanded: @1W%t#7W@ @6W%t#257K@ / @5W%t#7W@ @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_CPU_MR_MSG1_0012
Type	PLC alarm
ID	6
Alarm text	CPU maintenance required: @1W%t#7W@ @6W%t#257K@ / @5W%t#7W@ @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_CPU_MR_MSG1_0112
Type	PLC alarm
ID	7
Alarm text	CPU maintenance required: @1W%t#7W@ @6W%t#257K@ / @5W%t#7W@ @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_CPU_TMPERR_MSG_0013
Type	PLC alarm
ID	8

Totally Integrated Automation Portal		
Alarm text	Temporary CPU error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_RACK_MSG_0004	
Type	PLC alarm	
ID	9	
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_RACK_MSG_0104	
Type	PLC alarm	
ID	10	
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_DEVICE_MSG_0005	
Type	PLC alarm	
ID	11	
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_DEVICE_MSG_0105	
Type	PLC alarm	
ID	12	
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_IOSYSTEM_MSG_0006	
Type	PLC alarm	
ID	13	
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#276K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_IOSYSTEM_MSG_0106	
Type	PLC alarm	
ID	14	
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#276K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_MODUL_MSG_0003	
Type	PLC alarm	
ID	15	
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_MODUL_MSG_0103	
Type	PLC alarm	
ID	16	
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	

Totally Integrated Automation Portal		
Information only	False	
Name	SDIAG_ALCAT_SUBMODUL_MSG_0002	
Type	PLC alarm	
ID	17	
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_SUBMODUL_MSG_0102	
Type	PLC alarm	
ID	18	
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_CPU_OST_MSG_000D	
Type	PLC alarm	
ID	19	
Alarm text	CPU status message: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_CPU_OST_MSG_010D	
Type	PLC alarm	
ID	20	
Alarm text	CPU status message: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_PLC_MSG_00FF	
Type	PLC alarm	
ID	21	
Alarm text	PLC notification: @1W%t#7W@ @5W%t#7W@ @6W%t#256K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_PLC_MSG_01FF	
Type	PLC alarm	
ID	22	
Alarm text	PLC notification: @1W%t#7W@ @5W%t#7W@ @6W%t#256K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_CONFIG_REPORT_0029	
Type	PLC alarm	
ID	23	
Alarm text	Info: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_USER_MSG_0080	
Type	PLC alarm	
ID	24	
Alarm text	User message: @1W%t#2W@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_SECU_EV_MSG_005E	
Type	PLC alarm	
ID	25	

Totally Integrated Automation Portal		
Alarm text	Security event: @1W%t#7W@ @5W%t#7W@ @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_SECU_EV_INFO_005F	
Type	PLC alarm	
ID	26	
Alarm text	Security information: @1W%t#7W@ @5W%t#7W@ @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_SUB_ERR_MSG_001E	
Type	PLC alarm	
ID	27	
Alarm text	Error: @1W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_SUB_ERR_MSG_011E	
Type	PLC alarm	
ID	28	
Alarm text	Error: @1W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_SUB_MD_MSG_0021	
Type	PLC alarm	
ID	29	
Alarm text	Maintenance demanded: @1W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_SUB_MD_MSG_0121	
Type	PLC alarm	
ID	30	
Alarm text	Maintenance demanded: @1W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_SUB_MR_MSG_0024	
Type	PLC alarm	
ID	31	
Alarm text	Maintenance required: @1W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_SUB_MR_MSG_0124	
Type	PLC alarm	
ID	32	
Alarm text	Maintenance required: @1W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_CH_ERR_MSG_0015	
Type	PLC alarm	
ID	33	
Alarm text	Error: @1W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	

Totally Integrated Automation Portal		
Information only	True	
Name	SDIAG_ALCAT_CH_ERR_MSG_0115	
Type	PLC alarm	
ID	34	
Alarm text	Error: @1W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_CH_MD_MSG_0018	
Type	PLC alarm	
ID	35	
Alarm text	Maintenance demanded:@1W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_CH_MD_MSG_0118	
Type	PLC alarm	
ID	36	
Alarm text	Maintenance demanded:@1W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_CH_MR_MSG_001B	
Type	PLC alarm	
ID	37	
Alarm text	Maintenance required:@1W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_CH_MR_MSG_011B	
Type	PLC alarm	
ID	38	
Alarm text	Maintenance required:@1W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_CONFIG_INFO_0028	
Type	PLC alarm	
ID	39	
Alarm text	Info: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_CONFIG_INFO_0128	
Type	PLC alarm	
ID	40	
Alarm text	Info: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_ESUB_ERR_MSG_001F	
Type	PLC alarm	
ID	41	
Alarm text	Error: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_ESUB_ERR_MSG_011F	
Type	PLC alarm	

Totally Integrated Automation Portal		
ID	42	
Alarm text	Error: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_ESUB_MD_MSG_0022	
Type	PLC alarm	
ID	43	
Alarm text	Maintenance demanded: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_ESUB_MD_MSG_0122	
Type	PLC alarm	
ID	44	
Alarm text	Maintenance demanded: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_ESUB_MR_MSG_0025	
Type	PLC alarm	
ID	45	
Alarm text	Maintenance required: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_ESUB_MR_MSG_0125	
Type	PLC alarm	
ID	46	
Alarm text	Maintenance required: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_ECH_ERR_MSG_0016	
Type	PLC alarm	
ID	47	
Alarm text	Error: @1W%t#7W@ - @5W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_ECH_ERR_MSG_0116	
Type	PLC alarm	
ID	48	
Alarm text	Error: @1W%t#7W@ - @5W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_ECH_MD_MSG_0019	
Type	PLC alarm	
ID	49	
Alarm text	Maintenance demanded:@1W%t#7W@ - @5W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_ECH_MD_MSG_0119	
Type	PLC alarm	
ID	50	

Totally Integrated Automation Portal		
Alarm text	Maintenance demanded:@1W%t#7W@ - @5W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	
Name	SDIAG_ALCAT_ECH_MR_MSG_001C	
Type	PLC alarm	
ID	51	
Alarm text	Maintenance required:@1W%t#7W@ - @5W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	True	
Name	SDIAG_ALCAT_ECH_MR_MSG_011C	
Type	PLC alarm	
ID	52	
Alarm text	Maintenance required:@1W%t#7W@ - @5W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@	
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@	
Information only	False	

factoryio [CPU 1511-1 PN]

PLC alarm text lists

This folder is empty.

factoryio [CPU 1511-1 PN] / Local modules

factoryio [CPU 1511-1 PN]

factoryio

General\Project information

Name	factoryio	Author	Andrei
Comment		Rack	0
Slot	1		

General\Catalog information

Short designation	CPU 1511-1 PN	Description	CPU with display; work memory 150 KB code and 1 MB data; 60 ns bit operation time; 4-stage protection concept, technology functions: motion control, closed-loop control, counting & measuring; tracing; PROFINET IO controller, supports RT/IRT, performance upgrade PROFINET V2.3, 2 ports, I-device, MRP, MRPD, transport protocol TCP/IP, secure Open User Communication, S7 communication, Web server, DNS client, OPC UA server data access, constant bus cycle time, routing; Runtime options, firmware V2.1
Article number	6ES7 511-1AK01-0AB0	Firmware version	V2.1
	False		

General\Identification & Maintenance

Plant designation		Location identifier	
Installation date	2016-04-07 13:20:53.143	Additional information	

General\Checksums

Text lists	FA 70 E8 75 1D 5A 8E 29	Software	07 AE 60 E1 17 B3 C1 61
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Connection resources\

	Station resources - Reserved - Maximum	Station resources - Reserved - Configured	Station resources - Dynamic - Configured	Module resources - factoryio [CPU 1511-1 PN] - Configured
Maximum number of resources:		10	54	64
	Maximum	Configured	Configured	Configured
PG communication:	4	-	-	-
HMI communication:	4	2	0	2
S7 communication:	0	-	0	0
Open user communication:	0	-	0	0
Web communication:	2	-	-	-
Other communication:	-	-	0	0
Total resources used:		2	0	2
Available resources:		8	54	62

Overview of addresses\Overview of addresses\Overview of addresses

Inputs	True	Outputs	True
Address gaps	False	Slot	True

Totally Integrated Automation Portal							
Type	I	Addr. from	0	Addr. to	15	Module	AI 8xU//RTD/TC ST_1
PIP	None	OB	-	Device name	factoryio [CPU 1511-1 PN]	Device number	-
Size	16 Bytes	Master / IO-system	-	Rack	0	Slot	2
Type	I	Addr. from	16	Addr. to	31	Module	AI 8xU//RTD/TC ST_2
PIP	None	OB	-	Device name	factoryio [CPU 1511-1 PN]	Device number	-
Size	16 Bytes	Master / IO-system	-	Rack	0	Slot	3
Type	I	Addr. from	32	Addr. to	47	Module	AI 8xU//RTD/TC ST_3
PIP	None	OB	-	Device name	factoryio [CPU 1511-1 PN]	Device number	-
Size	16 Bytes	Master / IO-system	-	Rack	0	Slot	4
Type	I	Addr. from	48	Addr. to	63	Module	AI 8xU//RTD/TC ST_4
PIP	None	OB	-	Device name	factoryio [CPU 1511-1 PN]	Device number	-
Size	16 Bytes	Master / IO-system	-	Rack	0	Slot	5
Type	I	Addr. from	512	Addr. to	513	Module	DI 16x24VDC BA_1
PIP	None	OB	-	Device name	factoryio [CPU 1511-1 PN]	Device number	-
Size	2 Bytes	Master / IO-system	-	Rack	0	Slot	6

factoryio [CPU 1511-1 PN] / Local modules

AI 8xU//RTD/TC ST_1

AI 8xU//RTD/TC ST_1			
Name	AI 8xU//RTD/TC ST_1	Rack	0
Slot	2	Article number	6ES7 531-7KF00-0AB0
Short designation	AI 8xU//RTD/TC ST	Firmware version	V2.0

factoryio [CPU 1511-1 PN] / Local modules

AI 8xU//RTD/TC ST_2

AI 8xU//RTD/TC ST_2			
Name	AI 8xU//RTD/TC ST_2	Rack	0
Slot	3	Article number	6ES7 531-7KF00-0AB0
Short designation	AI 8xU//RTD/TC ST	Firmware version	V2.0

factoryio [CPU 1511-1 PN] / Local modules

AI 8xU//RTD/TC ST_3

AI 8xU//RTD/TC ST_3			
Name	AI 8xU//RTD/TC ST_3	Rack	0
Slot	4	Article number	6ES7 531-7KF00-0AB0
Short designation	AI 8xU//RTD/TC ST	Firmware version	V2.0

factoryio [CPU 1511-1 PN] / Local modules

AI 8xU//RTD/TC ST_4

AI 8xU//RTD/TC ST_4			
Name	AI 8xU//RTD/TC ST_4	Rack	0
Slot	5	Article number	6ES7 531-7KF00-0AB0
Short designation	AI 8xU//RTD/TC ST	Firmware version	V2.0

factoryio [CPU 1511-1 PN] / Local modules

DI 16x24VDC BA_1

DI 16x24VDC BA_1			
Name	DI 16x24VDC BA_1	Rack	0
Slot	6	Article number	6ES7 521-1BH10-0AA0
Short designation	DI 16x24VDC BA	Firmware version	V1.0